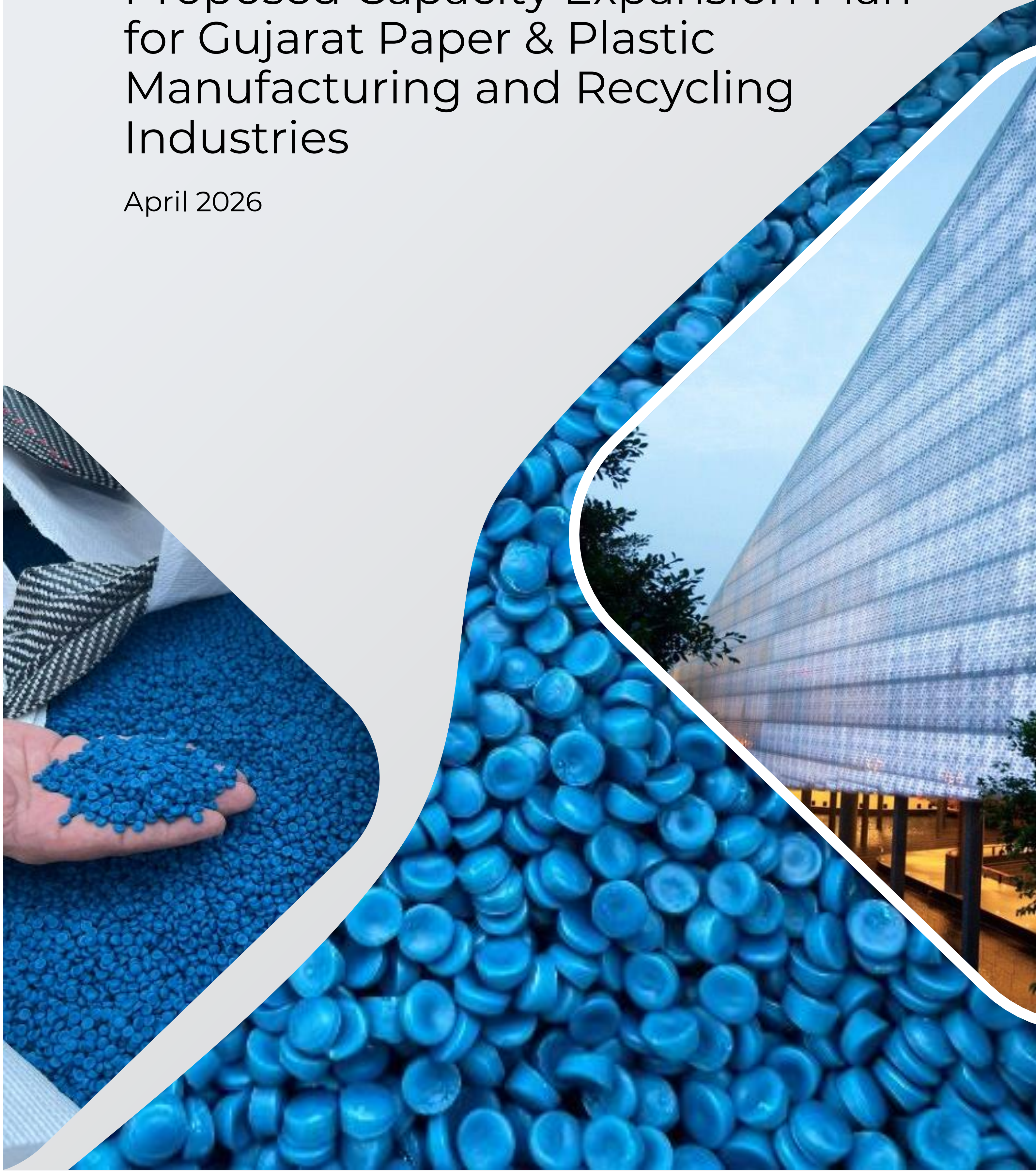
Techno-Economic Viability of Proposed Capacity Expansion Plan for Gujarat Paper & Plastic Manufacturing and Recycling Industries

April 2026



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Executive Summary

1.1 Project Overview

Gujarat Paper & Plastic Manufacturing and Recycling Industries LLC, established in 2019 in Abu Dhabi and presently operating in Umm Al Quwain, UAE, is a fully integrated and sustainable plastic recycling and manufacturing enterprise.

The Company specializes in converting post-consumer and industrial plastic waste into high-quality HDPE/LDPE granules, regrind plastics, off-grade materials, and finished plastic products such as shopping and garbage bags.

The existing plant operates with high efficiency (22 hours per day, 26 days per month). Through efficient utilization of advanced machinery and process optimization, the company produces recycled pellets, plastic films, and industrial plastics that cater to both domestic and regional markets. These operations play a crucial role in supporting the UAE's circular economy by reducing landfill dependency, minimizing plastic pollution, and promoting environmentally responsible manufacturing practices.

1.2 Business Case

To address the growing demand for sustainable, recyclable, and industrial-grade packaging materials across the UAE and GCC markets, the company proposes a significant capacity expansion initiative. The project involves scaling up production capacity by an additional 18,000 MT per annum, increasing the total installed capacity from 6,000 MT to 24,000 MT. This expansion is strategically aligned with regional infrastructure growth, rising industrial packaging requirements, and increasing regulatory emphasis on recycling and circular economy practices.

To support the expanded operations, the company plans to lease ~ 29,000 sq. ft. of adjoining industrial land, thereby doubling the existing operational footprint. The new facility will be dedicated to the manufacturing of PP/HDPE woven fabric and related industrial packaging products. The expanded infrastructure will allow for improved workflow efficiency, enhanced storage capacity, and optimized production layout.

1.3 Investment Ask and Returns Snapshot

The company proposes a capacity expansion project involving the installation of an additional 18,000 MT per annum woven packaging manufacturing line, increasing total installed capacity to 24,000 MT. The expansion will be executed on an adjoining 29,000 sq. ft. industrial plot and will incorporate advanced machinery from Lohia Corp.

Total Project Funding Structure:

- External Debt: AED 14.4 million
- Promoter Loan: AED 4.0 million
- Total Project Investment Considered: AED 18.4 million

The funding structure reflects a balanced capital mix with meaningful promoter commitment, demonstrating financial alignment and risk sharing.

Capacity & Revenue Scale-Up

- Existing Capacity: 6,000 MT
- Post-Expansion Capacity: 24,000 MT
- Steady-State Production: ~20,400 MT
- Revenue Growth: From AED 11.3 million (FY2025) to ~AED 76.4 million (FY2035)

Returns Snapshot

The projections indicate moderate cash flow pressure during the initial ramp-up phase (Years 1-2), primarily due to interest burden and partial capacity utilization. However, from 3rd Year onward, operating leverage improves materially, resulting in stabilized EBITDA margins, strengthening DSCR, and sustained positive free cash flow generation.

Return metrics are considered acceptable for capital-intensive industrial manufacturing projects within the UAE context, with adequate break-even cushion and resilience under moderate stress scenarios. Long-term financial sustainability is contingent upon disciplined execution, timely commissioning, stable raw material sourcing, and maintenance of contribution margins.

Project-Level Returns

Project IRR (Unlevered): ~18-20%

Payback Period: ~7 years

Break-Even Utilization: ~57%

Steady-State EBITDA: ~AED 5.5 - 7.4 million annually

The expansion results in a fourfold increase in installed capacity and significant revenue scalability driven by volume expansion and product diversification.

Risk Assessment

The project carries moderate operational and financial risk during the commissioning and ramp-up phase. Key risks include pricing volatility, raw material cost fluctuations, and capacity absorption timelines.

However:

- Machinery is sourced from a reputed vendor with local support
- Raw material procurement benefits from diversified supplier relationships and trading experience
- Break-even cushion provides operational safety margin
- Sensitivity analysis confirms viability under moderate stress scenarios

Regulatory environment remains favourable due to strong government support for recycling and sustainable packaging initiatives.

The proposed capacity expansion is strategically potential, technically fit, and financially viable. The project demonstrates attractive long-term returns, manageable leverage structure, and sustainable steady-state cash flow generation. While the first two years represent a structured ramp-up risk phase, the long-term economic fundamentals remain strong

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Project Rationale

The proposed expansion of *Gujarat Paper & Plastic Manufacturing and Recycling Industries LLC* is strategically aligned with national sustainability goals, evolving industrial demand, and the company's long-term growth vision. The project builds upon the strong foundation of existing operations and leverages technological excellence to position the company as a leading player in the UAE's green manufacturing ecosystem.

2.1 Global and Local Environmental Regulations

Alignment with UAE's Sustainability and Circular Economy Vision

The UAE generates ~ 900,000 tonnes of plastic waste annually, with less than 10% currently recycled through organized facilities. Under the UAE Vision 2031 and National Waste Management Strategy 2030, the government aims to divert 75% of waste from landfills and strengthen Extended Producer Responsibility (EPR) frameworks.

Gujarat Paper & Plastics' operations directly contribute to these objectives by transforming post-consumer and industrial waste into valuable products, reducing environmental impact, and fostering a circular manufacturing model.

2.2 Social and Economic Impact Assessment

Leveraging Strong Operational Base and Proven Efficiency

The existing plant operates with 22-hour daily uptime and over 80% utilization. This consistent performance showcases the company's strong process discipline, effective maintenance regime, and optimized manpower efficiency. The expansion will build on these strengths to achieve economies of scale and improve production efficiency across the value chain.

Responding to Market Demand for Woven Fabrics and Industrial Packaging

The UAE and GCC markets are witnessing rapid growth in demand for woven sacks, jumbo bags, and coated industrial fabrics, driven by construction, petrochemicals, and agriculture sectors. The regional woven packaging market is projected to grow at 8-10% CAGR over the next five years. The planned expansion will enable the company to capture a significant share of this growing segment through the installation of Lohia Corp's high-speed tape extrusion and circular loom systems, enhancing production capability by an additional 500 tonnes per month.

Enhancing Export Competitiveness and Market Reach

The company's location in Umm Al Quwain's industrial zone, with proximity to major seaports such as Jebel Ali and Sharjah, provides logistical advantages for exports across GCC, Africa, and South Asia. The addition of woven products and industrial packaging will open new export opportunities, potentially generating 20-25% of total sales from overseas markets within three years of expansion.

Strengthening Profitability through Cost-Effective Inputs

The company benefits from reliable access to industrial plastic waste streams - including blue drums, pipes, and films - at 20-30% lower input cost than virgin polymers. With the expansion, waste procurement partnerships will be extended to regional industries and waste management firms, ensuring stable raw material supply and improving gross margins by 3-5% over current levels.

Adoption of High-Quality Machinery and Process Integration

The expansion project leverages advanced, energy-efficient machinery from Lohia Corp, ensuring superior throughput, consistent quality, and lower maintenance downtime. Integrated process automation (PLC-based control, melt metering, and film thickness monitoring) will further reduce material losses and enhance productivity per unit of energy consumed.

Employment Generation and Economic Contribution

The expansion will need more than 3 times of current workforce, creating skilled and semi-skilled jobs in operations, quality control, and logistics. The project will contribute to local economic development, promote industrial sustainability, and support UAE's Make it in the Emirates initiative by localizing production of recyclable packaging materials.

Long-Term Strategic Fit

The project positions the company as a regional leader in sustainable polymer solutions, with a diversified product mix spanning granules, films, bags, and woven packaging. By combining recycling expertise with value-added manufacturing, Gujarat Paper & Plastics will enhance its market resilience, achieve higher capacity utilization, and strengthen its financial stability for future expansion or export-oriented partnerships.

2.3 Management Team

Sachin Shah - Founder & Managing Director

Sachin is a dynamic entrepreneur with over two decades of experience in plastic recycling, waste management, and manufacturing operations across the UAE and India.

With deep industry expertise and strong operational leadership, he has successfully built and scaled ventures in highly competitive and evolving markets. His entrepreneurial resilience was particularly evident during the global COVID-19 pandemic (2020-2022), where he navigated unprecedented disruptions with strategic agility, ensuring business continuity and sustained growth.

Sachin is the founder of Gujarat Paper & Plastic Manufacturing and Recycling Industries, established with a clear vision: to build a sustainable, technology-driven enterprise that transforms waste into high-value industrial products. Under his leadership, the company focuses on operational excellence, innovation in recycling processes, and creating measurable environmental impact while delivering commercial value. His experience of technology and operations is too deep which is primarily because of his ability to run machine at his own and being in charge of training to new staff.

Driven by a long-term perspective, Sachin combines sustainability, profitability, and scalable systems to create future-ready manufacturing ecosystems.

Management Team

The company is backed by a qualified and experienced management team that ensures strong functional oversight and seamless plant operations. The leadership team brings together diverse expertise across operations and finance, enabling disciplined execution, financial prudence, and operational efficiency. This integrated management approach strengthens governance, enhances decision-making, and supports sustainable growth.

Introduction to the Market

Plastic waste management has emerged as a critical environmental and economic priority for the UAE, driven by rapid urbanization, high per-capita plastic consumption, and ambitious national sustainability goals. As the country accelerates its transition toward a circular economy, plastic recycling is gaining strategic importance as both a waste-diversion solution and a value-creating industry.

Additionally, beyond just recycling, the United Arab Emirates (UAE) plastic granules market is poised for sustained long-term growth driven by macroeconomic fundamentals, progress in manufacturing and strategic diversification initiatives. Over the forecast horizon

3.1 Overview of the UAE and Regional (Middle East) Plastic Recycling Market

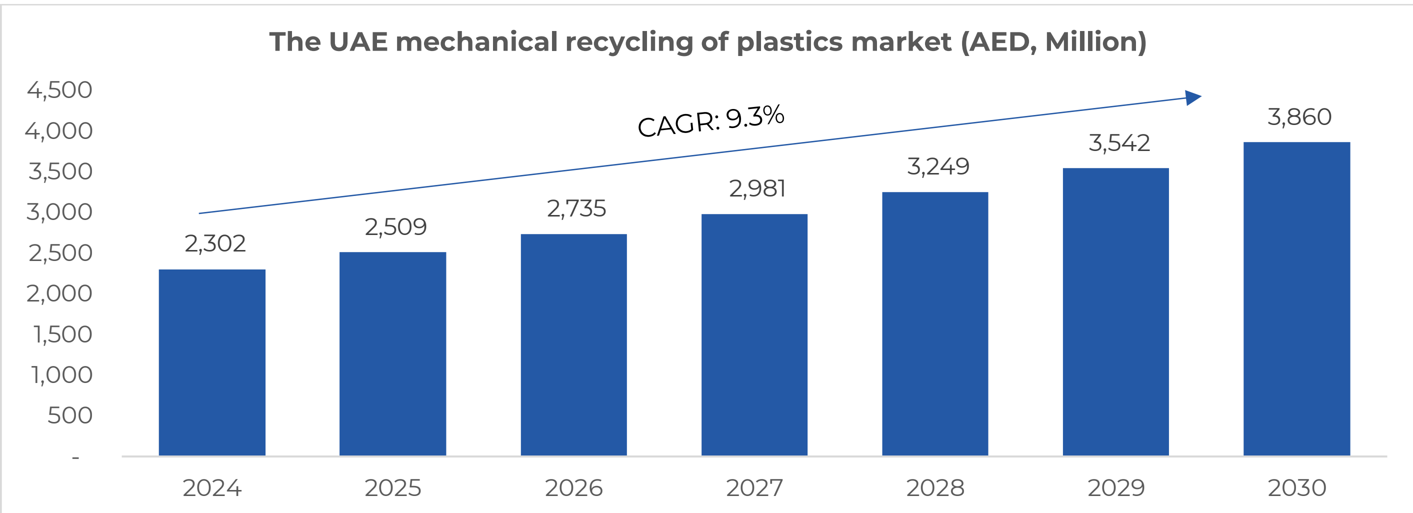
The plastic recycling and production industry is experiencing dynamic growth, driven by increasing environmental awareness, regulatory pressures, and the demand for sustainable materials. In the Middle East as well as globally, there is a shift towards circular economy models, with industries and governments emphasizing the reduction, reuse, and recycling of plastics. In the UAE, the sector is evolving rapidly, fuelled by government initiatives, investments in recycling infrastructure, and a growing market for recycled plastic products.

3.2 Market Analysis

The plastic recycling industry in the UAE and the broader Middle East is currently transitioning from a highly fragmented, small-scale, and informal system toward a more formal, technology-driven ecosystem. While still characterized by numerous small and medium players, the market is undergoing consolidation driven by government sustainability mandates and rising demand for recycled materials.

A. UAE Plastic Recycling Market Growth

The UAE mechanical recycling of plastics market generated approx. AED 2,302 million in 2024 and is projected to reach AED 3,860 million by 2030, reflecting a strong CAGR of 9.3% from 2025 to 2030. This growth is driven by regulatory push, circular economy initiatives, corporate ESG mandates, and increasing industrial demand for recycled polymers.



Conversion rate used 1 USD = 3.67 AED

In 2024, Polyethylene (PE) emerged as the largest revenue-generating segment and is expected to remain the most lucrative product category during the forecast period, registering the fastest growth.

PE dominates due to its extensive use in packaging, films, containers, and industrial applications, creating strong demand for recycled material substitution.

In 2020, the UAE plastic recycling market stood at ~ 0.84 million tonnes and forecasted to reach 1.44 million tonnes by 2030.

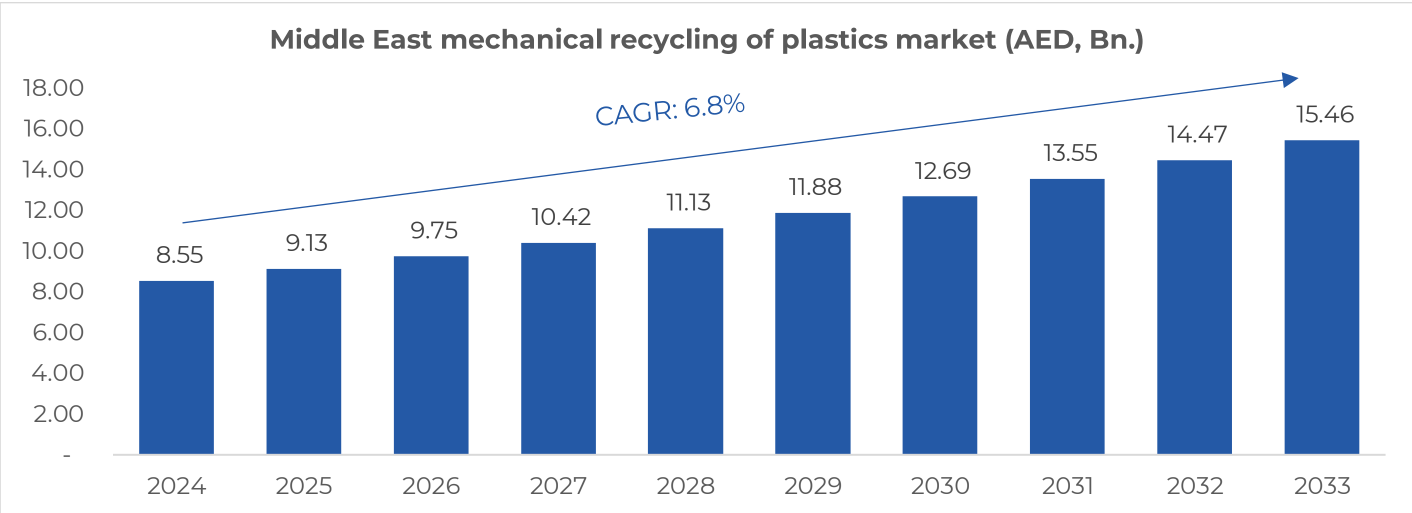
This indicates significant capacity expansion opportunities across the recycling value chain, particularly in mechanical recycling infrastructure and value-added recycled polymer manufacturing.

B. Middle East Regional Market Expansion Opportunity

In Saudi Arabia, municipal solid waste (MSW) generation is estimated at ~ 17 million tonnes annually, of which nearly 20% consists of plastic waste. However, only around 5% of this plastic waste is currently recycled, highlighting a significant gap between waste generation and recycling capacity.

This under-penetration presents a substantial opportunity to scale collection and recycling infrastructure across the KSA. More broadly, similar structural gaps exist across several GCC markets, indicating strong potential to expand recycling rates by multiples through investment in integrated waste management systems, advanced processing technologies, and formalized collection networks.

At the regional level, the Middle East mechanical recycling of plastics market was valued at ~ USD 2.33 billion in 2024 and is projected to reach around USD 4.20 billion by 2033, reflecting a steady CAGR of ~ 6.8%.



Across other GCC countries, although detailed public market data remains limited, regulatory reforms and sustainability-driven policy initiatives are accelerating, supported by high waste generation, rising consumption, urbanization, and industrial growth.

Infrastructure investments are also scaling, recently the joint venture between MVW Lechtenberg and Saudi Investment Recycling Company (SIRC) in Saudi Arabia targeting the processing of ~ 3 million tonnes of municipal solid waste annually.

3.3 Mega-Trends Driving Growth

The plastic recycling industry is undergoing structural transformation driven by regulatory reforms, sustainability imperatives, economic pressures, and technological advancements. What was previously considered a waste-management activity is now evolving into a strategic industrial sector aligned with circular economy principles. Governments across the UAE and GCC are tightening environmental regulations, global brands are mandating recycled content in packaging, and industries are actively seeking cost-efficient and sustainable raw material alternatives.

These long-term trends are not cyclical in nature; rather, they represent a structural shift in production and consumption patterns, creating sustained demand for recycling infrastructure and recycled plastic products.

Key mega-trends supporting growth in the plastic recycling sector include:

1. Regulatory Push: Policy as a Growth Catalyst

Regulatory frameworks across the UAE and broader GCC are accelerating demand for recycling and recycled content:

UAE Plastic Bans: The ban on single-use plastic shopping bags - effective January 1, 2024 - demonstrates credible regulatory enforcement against non-sustainable plastics. The ban's scope expands to plastic cups, plates, and cutlery from January 1, 2026, signalling progressive tightening of plastic product restrictions. This regulatory tightening elevates demand for recycled and sustainable packaging alternatives.

National Waste Management Agenda: The Ministry of Climate Change and Environment's "National Agenda for Integrated Waste Management" (Dec 2024) embeds circular economy principles into national strategy. It prioritizes waste reduction, recycling optimization, and sustainable material utilization - directly enhancing the structural demand for recycling capacity and higher-grade recycled resins.

Regional Momentum: Complementary regulatory efforts in the region - such as the Saudi Investment Recycling Company's MoU with Alliance to End Plastic Waste, targeting a 94% landfill diversion by 2035 - signal GCC-wide convergence on extended producer responsibility frameworks and landfill diversion commitments. Recycled beneficiation becomes economically and legally advantageous for packaging producers.

Regulatory policy is no longer speculative; it is a confirmed demand driver for recycled feedstocks and compliant packaging formats. This materially strengthens the increasing opportunities in recycling and downstream packaging.

2. Sustainability & ESG Imperatives

Demand drivers extend beyond regulation into corporate sustainability commitments:

- Global brands increasingly mandate ESG-compliant supply chains and recycled content in packaging to retain brand equity and market access.
- Industry initiatives (e.g., ISCC+ certification for circular polymers from integrated producers like Aramco/Total/SABIC) reflect the financial and reputational value of traceable recycled content.
- ESG requirements elevate recycled plastic from an environmental preference to a supply chain requirement, reducing buyer risk and improving price stability for recycled content.

3. Economic Drivers: Cost, Supply Security & Import Reduction

- **Oil Price Volatility:** Virgin plastics are tied to petroleum price cycles and import dependencies, creating cost volatility. Recycled plastics provide a cost-competitive hedge and predictable margin structures for packaging converters.
- **Import reduction:** GCC markets historically import significant volumes of virgin plastic resins and granules. Local recycling and value-added conversion mitigate tariff exposure, logistics risk, and foreign exchange pressures.

Domestic recycling projects exhibit revenue defensibility against commodity volatility and supply chain disruptions enhancing credit predictability.

4. Technology Evolution: Quality & Product Scope

- **Mechanical + Emerging Chemical Recycling:** Mechanical recycling remains the established backbone for polyolefins and PET recovery. Enhancements in sorting, washing, and decontamination technologies are improving output quality, enabling expanded use in packaging grades that were previously uncompetitive.
- **Catalysts for Hard-to-Recycle Streams:** Early commercialization of chemical recycling (e.g., pyrolysis, solvolysis) is broadening recovery scope within mixed and contaminated streams, supporting capacity scale-up.

Technological maturation reduces operational risk, supports higher yield and quality of recycled resins, and strengthens the long-term bankability of expansion investments.

5. Consumer & Market Demand Shifts

- **Consumer Awareness:** Growing public concern about plastic pollution translates into preference for recycled and recyclable packaging.
- **Packaging Market Profile:** The UAE's flexible packaging sector remains dominated by plastics (~65% share) with bags/pouches representing ~45.4% of formats. Regulatory pressure and consumer preference are increasing the adoption of mono-material and recyclable formats, aligning with the project's modernized production focus.

End-market demand for recycled packaging solutions is expanding on both corporate procurement and consumer fronts, reducing absorption risk for additional capacity.

Trend Category	Key drivers	Strategic Implication
Regulatory	UAE single-use bans + circular waste agenda	Structural increase in recycled resin demand
Sustainability	ESG adoption & certification requirements	Improved pricing power & demand stability
Economic	Cost arbitrage vs virgin plastics	Margin resilience + local supply advantage
Technology	Improved recycling quality/access	Higher utilization and product scope
Market	Consumer preference + packaging growth	Demand certainty for expanded capacity

3.4 Competitive Landscape

As an indication of regional feasibility of plastic recycling business in the region

The plastic recycling industry in the UAE and the broader Middle East is currently transitioning from a highly fragmented, small-scale, and informal system toward a more formal, technology-driven ecosystem. While still characterized by numerous small and medium players, the market is undergoing consolidation driven by government sustainability mandates and rising demand for recycled materials.

Umm Al Quwain has significance presence of plastic recycling units which indicates towards regional feasibility of plastic recycling. Over the years plastic recycling business

A. Shalimar Plastic Industries LLC, Umm Al Quwain

Shalimar is among the oldest players in the UAE plastic sector, with strong brand recognition and diversified product lines. Its long operating history and group backing provide financial and operational stability.

- Pioneer in plastic manufacturing, established in 1983
- Operates multiple divisions covering PE, PET, and PP products
- Part of Talal Group International, Dubai

B. Al-Ghori Plastic Recycling Factory LLC, Umm Al Quwain

Al-Ghori represents a major integrated recycler and granule supplier in the region. Its scale and export orientation make it a significant competitor in recycled raw materials.

- Established in 1996, part of Ghori Global Group
- Manufactures PP straps and recycled HDPE, LDPE, and PP granules
- Operates a large facility in Umm Al Quwain
- Serves export as well as domestic markets

C. Flax Plastic Manufacturing, Umm Dera, Umm Al Quwain

Flax Plastic focuses more on specialized and industrial plastic products. Its quality certifications and diversified customer base strengthen its market positioning.

- ISO 9001:2008 certified manufacturer
- Produces engineering, dairy, bakery, and industrial plastic products
- Supplies across the UAE and GCC

D. Fujairah Plastics Factory, Umm Al Quwain

Although located in Fujairah, this company is a major national-level competitor due to its scale, capacity, and regional market reach.

- Established in 1982
- Operates multiple manufacturing facilities
- Strong export presence
- Significant player in packaging and industrial plastics

E. Four Seasons FZE, Umm Al Quwain

Four Seasons operates mainly as a multi-material recycling and waste management company. Its diversified portfolio supports feedstock aggregation and recycling services rather than focused downstream manufacturing.

- Engaged in recycling of plastic scrap, metal scrap, e-waste, paper rolls, and polymers
- Provides waste disposal and environmental management services

Other Regional Players

No.	Company Name	Location	Main Activity	Materials Handled
1	Rose Polymers FZE	Sharjah	Recycling & Granule Manufacturing	HDPE, LDPE, PP, PVC
2	Planet Green Recycling	Dubai	Plastic Recycling & Waste Management	Mixed Plastics, HDPE, PET
3	Interpolymer (Harwal Group)	UAE	Large-Scale Polymer Recycling	PE, PP, PS
4	A to Z S Waste Recycling & Trading	Dubai / Sharjah	Scrap Processing & Regranulation	HDPE, LDPE, PP
5	SAN Plastic Waste Recycling	Sharjah	Recycling & Pelletizing	ABS, PS, PP, HDPE
6	Carbokene Polymers	Sharjah	Polymer Trading & Masterbatch	Recycled & Virgin Polymers
7	Qaamil Group	Sharjah	Recycling & Polymer Trading	Recycled PE, PP

The presence of multiple established players within Umm Al Quwain and across the UAE reflects the maturity and economic viability of the plastic recycling and manufacturing sector. The clustering of recycling and plastic processing units in key industrial zones indicates strong ecosystem development, availability of feedstock channels, established supply networks, and access to skilled technical resources.

Rather than indicating market saturation, the number of active participants demonstrates sustained demand, regulatory alignment, and commercial attractiveness of the industry. The sector shows characteristics of an increasingly organized and structured market, supported by:

- ✓ Established industrial infrastructure and regulatory oversight
- ✓ Growing integration between recycling and downstream manufacturing
- ✓ Expanding domestic and regional demand for recycled polymers and plastic products
- ✓ Rising compliance with environmental and quality standards
- ✓ Strong linkages between waste collection, processing, and end-use industries

An organized industry structure enhances operational efficiency, strengthens supply chain reliability, and promotes healthy competition, which in turn supports innovation and capacity expansion. The development of this ecosystem reduces entry barriers related to feedstock sourcing and market access while reinforcing long-term sustainability of operations.

Overall, the competitive landscape in Umm Al Quwain and the broader UAE reflects a high-potential and structurally supported industry, where continued growth in recycling and value-added plastic manufacturing is both commercially and strategically aligned with national sustainability objectives.

3.5 Target Customers Segments

Currently the Company primarily serves the packaging industry, which represents the majority of its customer base. It also supports the construction and consumer goods sectors by supplying specialized materials used in packaging.

Packaging Industry

- Used for flexible packaging, cartons, labels, and protective films
- Enhances product protection, shelf life, and transportation safety
- Supports branding, traceability, and regulatory compliance

Construction Industry

- Currently serves very limited offering in construction directly, primarily serves films, sheets to cover hardware etc.

Consumer Goods Industry

- Supplies packaging components and protective layers for finished products
- Supplies plastic granules of food grade for water bottles
- Improves product appearance, usability, and shelf presentation
- Supports sustainable packaging and eco-friendly initiatives

Potential Market Segments - Demand Mapping & Revenue Diversification

The proposed expansion into woven polypropylene (PP) fabrics, industrial sacks, recycled plastic granules, and value-added packaging products enables the Company to address multiple diversified end-use industries across the UAE, broader GCC region, and select export markets.

The identified market segments are largely essential-use categories with recurring consumption characteristics, thereby supporting revenue stability and mitigating concentration risk. The following demand mapping outlines segment drivers, revenue potential, and associated risk considerations.

Cement & Construction Industry Packaging

The cement and construction materials segment represents a core demand vertical for woven polypropylene sacks in the GCC region. Infrastructure development programs across the UAE and neighbouring economies, supported by public investments and industrial expansion, continue to generate sustained demand for cement and other construction materials. These products require high-strength woven sacks in standard 25 kg and 50 kg formats with enhanced moisture resistance and load-bearing capacity.

Product Fit	
50kg cement bags	Valve-type sacks
Laminated woven sacks	Gusseted industrial bags

Demand Characteristics	
High volume, stable demand	Long-term supply contracts common
Quality standardization critical	Moderate pricing pressure

High volume, standardized product specifications, and long-term procurement contracts with cement manufacturers and construction material suppliers characterize demand in this segment. Pricing tends to be competitive; however, economies of scale and laminated or printed variants offer opportunities to enhance margins. The essential and infrastructure-linked nature of this sector supports baseline capacity utilization and relatively low cyclicality risk. As such, this segment is expected to provide a stable revenue anchor for the expanded operations.

Fertilizer, Agro-Products & Food Grain Packaging

The fertilizer and agro-products segment presents structurally stable demand driven by food security initiatives, agricultural imports, and regional trading activities. The GCC countries rely significantly on imported food grains, rice, sugar, pulses, and animal feed, all of which require durable woven sacks with UV stabilization and moisture protection.

Product Fit	
25kg and 50kg agricultural bags	Laminated fertilizer sacks
UV-stabilized agro packaging	Printed branding sacks

Demand Drivers	
Food security initiatives	Agricultural imports
Seasonal but recurring demand	Export shipments to Africa and South Asia

In addition, fertilizer distribution within the region necessitates laminated and printed sacks capable of withstanding storage and transport stress. Demand in this segment is moderately seasonal but recurring, with limited sensitivity to macroeconomic cycles due to its linkage to essential commodities.

Custom printing and branding features support moderate margin realization. This segment enhances portfolio stability and provides balanced exposure to both domestic distribution and re-export trade.

Polymer & Chemical Industry Packaging

The UAE functions as a major petrochemical trading and redistribution hub, creating consistent demand for industrial-grade woven packaging solutions. Polymer granule manufacturers, chemical distributors, and specialty compound suppliers require high-tensile, moisture-resistant sacks that comply with export handling standards.

Product Fit		
25kg polymer bags	Chemical sacks	Export-grade printed woven packaging

Demand Characteristics		
B2B industrial contracts	Strict quality compliance	Export-oriented packaging

This segment typically operates under B2B contractual arrangements with strict quality parameters and repeats order cycles. Compared to cement packaging, margin realization is relatively higher due to performance requirements and lower commoditization. However, quality compliance and product consistency are critical success factors. Exposure to this segment strengthens the Company's industrial positioning and enhances revenue quality through repeat institutional buyers.

FMCG, Retail & Logistics Packaging

The FMCG and retail segment includes applications such as garbage bags, shopping bags, courier bags, and flexible plastic packaging. Demand in this segment is driven by population growth, organized retail expansion, tourism activity, and increasing e-commerce penetration across the UAE.

Product Fit	
Shopping bags (HDPE/LDPE blend)	Garbage bags (100% recycled LDPE)
Courier bags	Printed carry bags

Demand Characteristics	
High repeat consumption	Price-sensitive
Strong regulatory oversight (single-use bans)	

While this segment is relatively price-sensitive and competitive, consumption patterns are frequent and recurring. The Company's integration into recycled LDPE and HDPE granule production provides cost advantage and regulatory compliance in light of tightening environmental norms. However, regulatory developments concerning single-use plastics require continuous product adaptation. This segment contributes to volume throughput and working capital rotation efficiency.

Industrial Bulk Packaging (FIBC / Jumbo Bags)

Flexible Intermediate Bulk Containers (FIBCs), commonly known as jumbo bags, serve bulk handling industries including cement exports, petrochemicals, sand, minerals, and agricultural commodities. These bags typically range from 500 kg to 2,000 kg capacity and require higher technical specifications and load certification standards.

Product Fit		
1-ton jumbo bags	Liner-fitted FIBCs	UV-treated bulk bags

Margin Profile		
Higher than standard sacks	Export-oriented premium	Quality certification required

The segment offers superior margin potential relative to standard woven sacks, albeit with higher entry barriers related to testing, compliance, and quality assurance. Demand is driven by export-oriented bulk commodity movement. While working capital requirements may be moderately higher due to customization and quality processes, this segment enhances revenue diversification and value addition within the woven fabric product line.

Export Market Opportunities (GCC, East Africa & South Asia)

The UAE's strategic geographic location and logistics infrastructure enable access to neighbouring GCC markets as well as East Africa and South Asia. Countries such as Oman, Saudi Arabia, Bahrain, Kenya, Tanzania, and Bangladesh present growing demand for cost-effective woven packaging solutions.

Target Markets		
Oman	Bahrain	Saudi Arabia
Kenya	Tanzania	

Strategic Benefit		
Geographic diversification	Currency diversification	Margin arbitrage in select corridors

Export exposure provides geographic diversification and potential margin arbitrage. However, it introduces foreign exchange exposure, counterparty credit assessment requirements, and trade regulation considerations. The Company's trading experience and regional supply linkages partially mitigate these risks. Export markets are expected to complement domestic sales rather than substitute them, thereby strengthening revenue resilience.

ESG-Compliant & Recycled Content Packaging

Sustainability-driven packaging solutions represent an emerging growth segment supported by regulatory enforcement and corporate ESG commitments. Extended Producer Responsibility (EPR) frameworks and recycled-content mandates are increasingly influencing procurement policies of institutional buyers.

The Company's backward integration into recycled granule production positions it favourably to supply certified recycled-content woven sacks and plastic packaging products. This segment offers potential pricing premiums and long-term growth visibility as sustainability norms tighten. Demand is expected to accelerate over the medium term, supporting margin expansion and enhancing the Company's environmental compliance profile.

Strategic Opportunity for Gujarat Paper & Plastics

The Company operates in structurally essential and high-consumption end-use segments, primarily within industrial and flexible packaging markets. Its current exposure to packaging, cement, agro-products, polymer, and consumer goods sectors provides a stable revenue base anchored in recurring demand and essential commodities. With the proposed expansion into woven polypropylene (PP) fabrics, industrial sacks, recycled granules, FIBCs (jumbo bags), and value-added packaging solutions, the Company is positioned to significantly broaden its addressable market across the UAE, wider GCC, East Africa, and South Asia.

The expansion strategically shifts the Company from a primarily packaging-focused supplier to a diversified industrial packaging and materials solutions provider. This enhances revenue visibility, improves margin mix through value-added products, enables geographic diversification, and reduces concentration risk. Importantly, the target segments are linked to infrastructure, food security, petrochemicals, retail consumption, bulk exports, and sustainability mandates - all of which demonstrate structural growth drivers rather than cyclical demand patterns.

Accordingly, the post-expansion business model is expected to deliver stronger capacity utilization, improved economies of scale, enhanced pricing power in specialized segments, and greater resilience across economic cycles.

Business Model Analysis: Evaluation of Business Operations

Gujarat Paper & Plastic Industries operates a full-cycle recycling and manufacturing model, starting from plastic waste procurement to finished bag products and generating revenue from multiple streams. The company also purchases ready-to-process plastic pallets to meet additional demand or specific requirements that cannot be satisfied through recycling alone or require a specific grade of plastic pallets. Future expansion focuses on capacity enhancement, new product development, technological automation, and regional market growth, aligning with sustainability trends and increasing demand for plastic recycling.

The Company is also engaged in trading however in this report the entire focus is on plastic recycling and manufacturing capabilities only considering trading business as additional opportunity not a core business activity.

4.1 Business Model Overview

Gujarat Paper & Plastic Industries operates an integrated recycling and manufacturing platform that spans the complete value chain - from plastic waste sourcing to the production of finished bag products - enabling the Company to capture value at multiple stages and generate diversified revenue streams.

Looking ahead, the Company's expansion roadmap is focused on capacity augmentation, introduction of new product categories, technological upgradation through automation, and geographic market expansion. These strategic priorities are aligned with accelerating sustainability mandates and the growing institutional preference for recycled plastic solutions, positioning the Company to capitalize on structurally expanding demand.

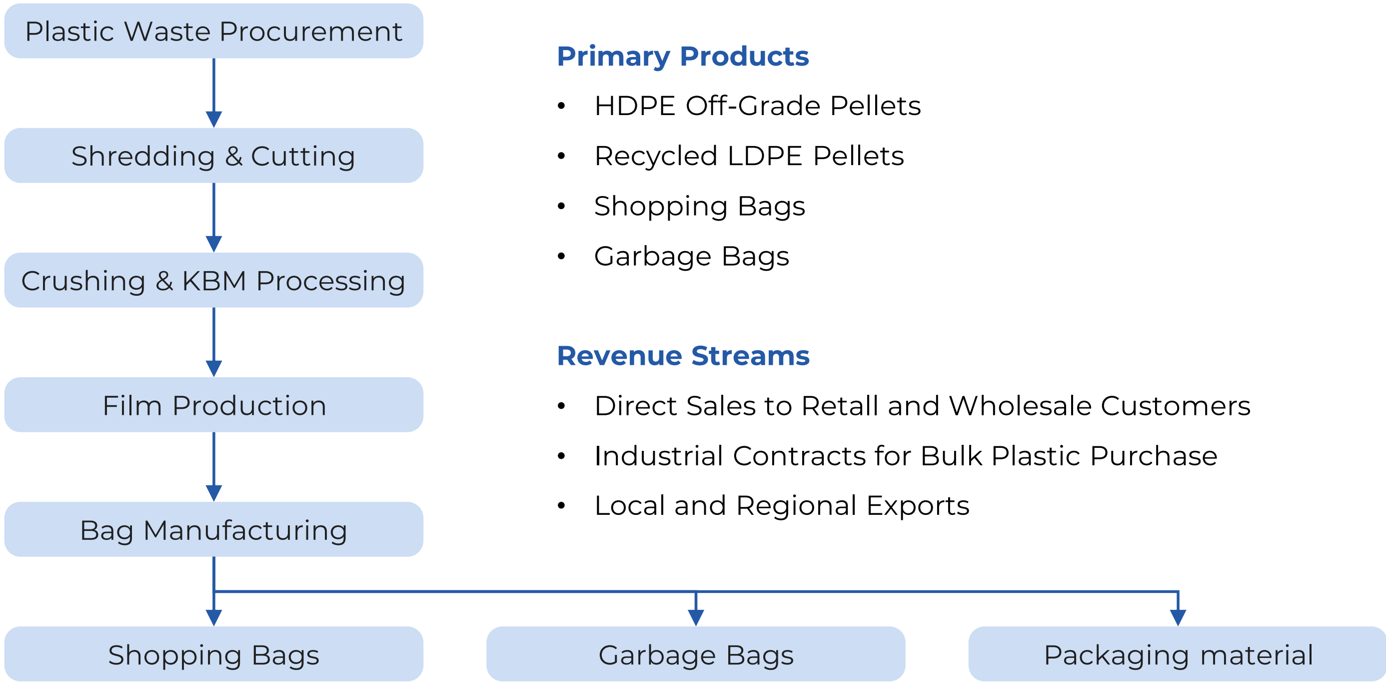
Leveraging its longstanding presence in the trading segment, the Company has developed strong relationships with end customers and plastic distributors. This trading experience has not only contributed meaningfully to revenue generation but has also provided management with deep insights into pricing behaviour, procurement cycles, customer specifications, and emerging demand trends. However, from TEV perspective, the capacity upgrade evaluation remains specific to existing recycling capabilities and impact of new capacity addition only.

While the leadership team possesses substantial industry knowledge and commercial acumen, the current operational framework remains largely conventional and requires modernization to meet evolving quality, efficiency, and scalability expectations. Management believes that by upgrading infrastructure, introducing process automation, and strengthening operational systems - while simultaneously leveraging its market intelligence from the trading business - the Company can unlock significant improvements in throughput, sales realization, and margin performance.

This section provides a comprehensive technical evaluation of the Company's existing recycling and manufacturing operations including

- Technical evaluation of the Company's current recycling and manufacturing operations
- Assessment of feedstock procurement mechanisms and sourcing structure
- Review of processing technology and production systems
- Review of capital deployment efficiency
- Evaluation of overall readiness to support the proposed expansion

The end-to-end business model focuses on sourcing plastic waste followed by systematic sorting, cleaning, and processing to remove contaminants and segregate materials by polymer type. The collected waste is then shredded, washed, dried, and recycled through extrusion and pelletizing to produce high-quality recycled plastic granules. These granules are further processed using film extrusion, printing, cutting, and sealing technologies to manufacture plastic films, bags, and other packaging materials



Primary Products

- HDPE Off-Grade Pellets
- Recycled LDPE Pellets
- Shopping Bags
- Garbage Bags

Revenue Streams

- Direct Sales to Retall and Wholesale Customers
- Industrial Contracts for Bulk Plastic Purchase
- Local and Regional Exports

4.2 Current Operations Assessment

Plastic Waste Procurement

Gujarat Paper & Plastic Industries currently maintains a multi-source procurement model to ensure consistent feedstock availability for recycling. The key sourcing channels include:

- **Industrial plastic scrap suppliers** (pipe manufacturing units, packaging industries, sheet and film manufacturers)
- **Post-consumer waste streams** collected through local aggregators and traders (Currently not procuring municipality waste due to limited installed capacity and space requirement for cleaning and segregation)

Strategic Objective: Ensure a continuous, cost-efficient supply of recyclable plastics to minimize feedstock shortages and maintain stable production utilization.

Quality Segregation & Volume Assurance:

- Incoming materials are classified into HDPE, LDPE, PP, mixed plastics, and other sub-grades
- Segregation is primarily manual, supported by basic inspection protocols
- While the process remains traditional, experienced operators contribute to consistent input quality control

Image 1: Industrial Waste of high-quality plastic procured from Pipe Industries, however, rejected by their client

Image 1A: Industrial Waste of high-quality plastic procured from various industries

4.3 Process Technology and Product Mix Assessment

Gujarat Plastic operates a conventional yet well-structured recycling line, supported by machinery that meets current-scale production needs. Although the systems are operationally adequate, there exists a significant opportunity for modernization to increase throughput, reduce operational losses, and improve energy efficiency.

Table: Existing Machinery & Equipment at Gujarat Plastic

Sr. No.	Process Stage	Equipment Name	Manufacturer / Model	Primary Function	Key Technical Features	Operational Status
1	Shredding & Size Reduction	Single-Shaft Shredding Machine	Wide Sky	Primary size reduction of plastic waste (woven sacks, films, lumps)	Heavy-duty single shaft design, energy-efficient drivetrain, continuous feed stability, compatible with PP, HDPE, LDPE	Fully Operational
2	Manual Pre-Cutting	Heavy-Duty Plastic Cutter	Dewalt 20V MAX Plastic Tubing Cutter	Cutting oversized and high-density plastic into manageable pieces	Cuts up to 2 inches thickness, burr-free precision cutting, automatic/manual modes, LED illumination	Fully Operational
3	Crushing & Pelletizing	Extrusion Pelletizing System	KBM Single-Screw Extruder	Conversion of shredded plastic into recycled pellets	High-efficiency single-screw extruder, degassing unit, temperature-controlled heaters, strand cutting pelletizer	Fully Operational
4	Pellet Cooling & Collection	Pellet Collecting & Cooling Unit	Integrated with KBM System	Cooling, solidifying and collecting pellets	Uniform cooling system, stable granule formation, moisture control	Fully Operational
5	Film Production	Blown Film Extrusion System	Five-Star Engineers	Production of recycled HDPE/LDPE film rolls	Precision flat die, uniform melt distribution, chilled air-cooling system, high tensile film output	Fully Operational

Sr. No.	Process Stage	Equipment Name	Manufacturer / Model	Primary Function	Key Technical Features	Operational Status
6	Film Production	Blown Film Extrusion System	King Hsing	Production of plastic films for bag manufacturing	Advanced blown film technology, roller cooling, thickness consistency, suitable for blended recycled materials	Fully Operational
7	Bag Conversion	Bag Cutting & Sealing Unit (Integrated Line)	In-house integrated system (King Hsing)	Conversion of films into shopping and garbage bags	Heat sealing, consistent edge finishing, adaptable to HDPE/LDPE blends	Fully Operational
8	Material Handling	Industrial Forklift	Lonking 30	Internal transportation of raw materials and finished goods	High load capacity, industrial manoeuvrability, fuel-efficient operation	Fully Operating

Processing Workflow

Shredding & Cutting

Machinery Deployed:

- Wide Sky Single-Shaft Shredding Machine
- Dewalt Heavy-Duty Plastic Cutter

Technical Assessment of the Wide Sky Shredder: The Wide Sky shredding unit forms the backbone of the recycling operations, serving as the primary shredder. It is fully operational and well-suited to the current plant size.

Design Strengths:

- Heavy-duty single-shaft system ensures continuous feed stability
- Efficient construction minimizing mechanical stress
- Energy-efficient drivetrain
- Compatible across polymer types: PP, HDPE, LDPE

Performance & Operational Fit

- Fully operational and aligned with current plant capacity
- Processes woven sacks, plastic films, rigid lumps, and mixed industrial scrap
- Compatible with core polymer streams: PP, HDPE, LDPE
- Ensures stable feed into downstream extrusion systems

Engineering Strengths

- Heavy-duty single-shaft configuration for consistent material intake
- Mechanical construction reducing vibration and wear
- Energy-efficient drivetrain supporting cost control
- Designed for continuous industrial-duty cycles

Operational Impact

- High uptime and dependable throughput
- Minimal material degradation during shredding
- Reduced mechanical stress on downstream equipment
- Low maintenance complexity and manageable servicing

Overall, the shredder is well-suited to the present production volume and contributes significantly to operational stability.

Technical Assessment of the Dewalt Cutter

The site also includes a **Dewalt Cutter,** which is a powerful tool designed to cut heavy plastic into smaller pieces efficiently. Specifically, the model on the site is the 20V MAX Plastic Tubing Cutter model, offering clean, burr-free cutting for plastic materials up to 2 inches thick.

The Dewalt cutter complements the shredding line by processing oversized and high-density plastic pieces:

- **Model Reviewed:** 20V MAX Plastic Tubing Cutter
- **Cutting Capability:** Up to 2 inches in thickness

Performance & Operational Fit: Operationally fit for smoother end-to-end workflow

Image 2: Wide Sky Shredding Machine is the key shredder in the Recycling Plant

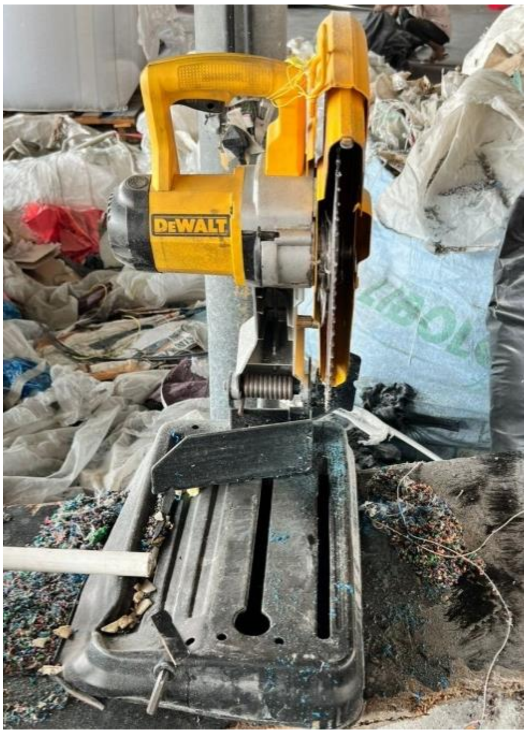


Image 3: Dewalt Cutter with blade to cut heavy plastic into small pieces

Image 4: Shredded Industrial Waste, which will be sent to the next step for pelletizing

Crushing & Pelletizing:

The Company deploys a high-efficiency KBM extrusion pelletizing system as its core pellet production platform, playing a critical role in transforming shredded industrial plastic waste into commercially usable recycled pellets. The system is engineered to manufacture multiple pellet grades, supporting diversified product output requirements. This Crushing & Palletizing section includes:

KBM Extrusion Pelletizing System

- Pellet Collecting & Cooling Unit

Performance & Operational Fit

- Converts shredded plastic into marketable recycled pellets
- Throughput aligned with current plant scale
- Supports multi-grade production mix requirements
- Maintains stable melt flow under continuous operation

Engineering Strengths

- High-efficiency single-screw extruder ensuring consistent melt quality
- Strand-cut pelletizer producing uniform pellet dimensions
- Integrated degassing system removing moisture and volatile contaminants
- Temperature-controlled heating zones ensuring melt homogeneity
- Energy-optimized system design

Operational Impact

- Consistent pellet quality suitable for industrial-grade applications
- Reduced product defects (air entrapment, structural inconsistencies)
- Stable long-duration performance with minimal mechanical interruption

- Competitive per-kg processing economics

Strategic Assessment

- Primary value-conversion engine of the plant
- Strong alignment with diversified recycled polymer strategy
- Demonstrates efficient capital deployment relative to output quality
- Technically capable of supporting phased capacity expansion and product portfolio scaling

Pellet Categories Produced

- HDPE off-grade pellets
- Recycled LDPE pellets
- PPE-grade industrial pellets

Image 5-10: KBM extrusion pelletizing system

Image 11: Pellets Collecting Unit

Image 12: Pellets from KBM machines produced during the site visit

Material Handling: Lonking 30 forklift

The site employs a Lonking 30 forklift for material handling and moving materials within and around the factory premises. It is well-suited for transporting raw materials like plastic pellets, shredded waste, and finished goods like plastic films and bags across various production stages.

Image 13: Lionking material handling forklift

4.3 Product Development

Film Production

Following the upstream recycling processes-shredding, cutting, crushing, and pelletizing the Company converts its in-house recycled pellets into value-added finished film and bag products. This vertical integration enhances margin capture, reduces external sourcing dependency, and strengthens the company's competitive positioning in the UAE and GCC packaging market. Additionally on demand basis the Company also use directly purchased ready to use pallets for film manufacturing.

Film Production Unit

Machinery Utilized:

- Five-Star Engineers Blown Film Extrusion System
- King Hsing Blown Film Extrusion System

The film production unit is equipped with blown film extrusion systems from Five Star Engineers and King Hsing, two manufacturers widely recognized for delivering reliable, high-precision plastic film production machinery. These systems are engineered to produce industrial-grade films with the performance characteristics required for downstream bag manufacturing.

Both machines utilize advanced blown film extrusion technology incorporating precision flat dies, enabling uniform melt distribution and consistent film thickness. Integrated cooling mechanisms, including high-efficiency chilled air and roller systems, ensure rapid film solidification, enhanced mechanical strength, and improved clarity.

The combined output capacity of the Five Star Engineers and King Hsing units provides with a dependable and continuous supply of high-quality plastic films. This ensures operational continuity in the bag-manufacturing division, enhances production efficiency.

Technical Overview & Assessment

- **Advanced Blown Film Technology:**
  Both machines employ precision flat dies to ensure uniform melt distribution, enabling:
  - Consistent film thickness
  - High tensile strength
  - Improved clarity and surface finish
- **Cooling & Solidification Systems:**
  High-capacity chilled air and roller-cooling mechanisms ensure rapid solidification. This contributes to:
  - Reduced defect rates
  - Enhanced mechanical properties
  - Higher line throughput

Operational Fit: The machinery configuration is well-aligned with the Company's recycled feedstock profile, ensuring efficient and stable processing of HDPE, LDPE, and blended film-grade materials.

Strategic Value Contribution

Film production represents a critical mid-stream integration point between pelletizing and final bag manufacturing, strengthening vertical control across the value chain. By internalizing film production, the Company achieves:

- Elimination of external supplier dependency, reducing procurement risk and supply volatility
- Enhanced input quality control for downstream bag manufacturing, ensuring consistency in thickness, strength, and finish
- Lower cost-per-unit economics, improving contribution margins through backward integration
- Improved market agility, enabling faster product changeovers and customization across bag types
- Collectively, in-house film manufacturing enhances operational control, margin resilience, and competitive responsiveness.

Images 14-17: Five-Star Engineers and King Hsing machines for Plastic Film Production

Bag Manufacturing:

The Company operates a fully integrated bag manufacturing unit that converts in-house-produced plastic films into two major product categories: **Shopping Bags and Garbage Bags**. This downstream production capability enhances value addition, stabilizes margins, and enables the company to serve both **retail and commercial markets** and **industrial waste-management sectors**.

Product Output Supported by Installed Machinery

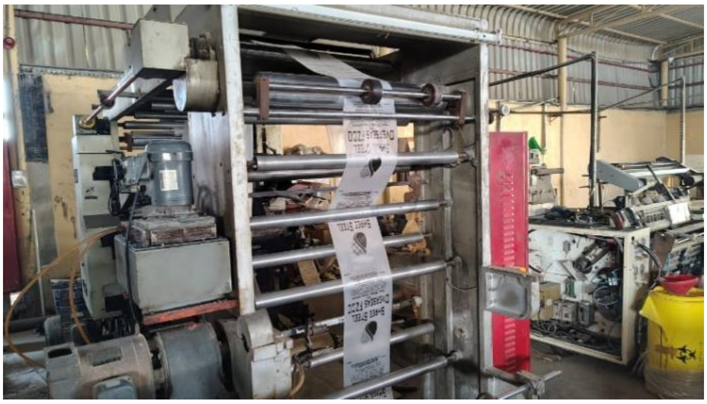
Product Category	Raw Material Source	Production Stage
HDPE Off-Grade Pellets	Industrial waste	KBM Extrusion Line
Recycled LDPE Pellets	Industrial waste	KBM Extrusion Line
PPE-Grade Industrial Pellets	Mixed industrial waste	KBM Extrusion Line
HDPE/LDPE Blown Film	In-house recycled pellets	Five-Star & King Hsing Lines
Shopping Bags (75% HDPE / 25% LDPE)	In-house film production	Bag Conversion Unit
Garbage Bags (100% recycled LDPE)	In-house film production	Bag Conversion Unit

Shopping/ Packaging Bags:

Manufactured using a 75% HDPE and 25% LDPE blend, the shopping bags combine structural strength with flexibility. HDPE provides rigidity, load-bearing capacity, and puncture resistance, while LDPE enhances stretchability, reduces brittleness, and improves surface finish. The result is a durable, retail-ready product with strong branding appeal and competitive production costs due to in-house processing. This line is well-positioned within the UAE's retail and grocery sectors, where reliability and consistent quality are essential.

Garbage Bags:

Produced from 100% recycled LDPE pellets, garbage bags are designed for flexibility, tear resistance, and moisture durability. The product caters to high-volume, recurring demand across waste management, hospitality, commercial facilities, and municipal sectors. This category provides stable revenue streams supported by predictable order cycles and repeat business.




Images 18: Bags printing machine

4.4 Revenue Model Analysis

Revenue Streams

The Company operates a **diversified revenue model** spanning manufacturing, industrial supply contracts, export operations, and a high-value trading division. This multi-source income structure enhances financial stability and reduces market dependency-an important factor in Technical and Economic Viability (TEV) assessment.

Revenue from Manufacturing Operations

Direct Sales - Domestic Market

The company earns revenue through domestic sales of its plastic films, bags, and recycled materials. Key customer segments include:

- Retailers and wholesalers
- Packaging vendors
- Local plastic and paper distribution agents

This revenue stream benefits from strong and stable UAE consumption, especially in FMCG, retail, hospitality, and general merchandise sectors. Established supply relationships ensure reliable cash inflows and repeat orders.

Industrial Supply Contracts

The Company supplies recycled pellets, semi-finished films, and material blends to industrial converters and manufacturers through long-term B2B contracts. These agreements provide:

- Predictable, high-volume monthly orders
- Structured payment cycles
- Lower customer acquisition and marketing costs

Export Sales (GCC & East Africa)

The company claims to leverages the UAE's geographic advantage and efficient port infrastructure to serve regional markets across:

- GCC countries
- East Africa
- Select Asian markets

Trading Division (Significant Revenue Contributor, however not considered for financial feasibility analysis)

Besides manufacturing, the Company operates a **substantial trading business**, which contributes a significant share of total annual revenue. Trading activities include sourcing and resale of:

- HDPE, LDPE, PP pellets (virgin and recycled)
- Imported and repackaged packaging products
- Supplementary raw materials required by downstream converters

4.5 SWOT Analysis

The SWOT analysis presents a structured evaluation of the Company's strategic position within the recycling and manufacturing sector. The Company benefits from an integrated recycling-to-manufacturing value chain, strong trading capabilities that provide real-time market intelligence, and reliable operational performance supported by high machine uptime.

These strengths, combined with growing export demand across GCC and emerging markets, as well as opportunities for capacity expansion and product diversification, create a solid platform for scalable growth.

At the same time, the assessment recognizes external pressures such as pricing competition, feedstock supply variability, regulatory shifts, and the risk of technology obsolescence, all of which require proactive strategic planning to sustain long-term competitiveness and margin resilience.

Strengths	Weaknesses
• Integrated recycling-to-manufacturing value chain enabling cost control and margin optimization • Strong trading division providing market intelligence, cash-flow stability, and supplier/customer negotiation leverage • High machine uptime and operational reliability • Cost-effective feedstock access from post-industrial and off-grade polymer markets • Established vendor and customer network ensuring sourcing stability and repeat sales	• Existing operational setup with limited automation and capacity constraints • Limited product diversification, with scope to expand into higher-margin segments (due to technology limitations) • Manual handling at key processing stages affecting consistency and throughput during peak demand
Opportunities	**Threats**
• Growing demand in GCC, East Africa, and South Asia driven by sustainability regulations and UAE manufacturing advantages • Capacity expansion and modernization to improve yield, reduce wastage, and enhance profitability • New product development opportunities (e.g., courier bags, industrial liners, agricultural and biodegradable films) • Leveraging trading insights for informed CAPEX decisions and strategic product planning	• Pricing competition from regional recyclers and low-cost imports impacting commodity margins • Feedstock supply variability due to economic cycles or regulatory changes • Regulatory shifts in plastic recycling policies demanding operational adjustments • Risk of technology obsolescence if modernization is delayed

Evaluation of Proposed Technology and Capacity Expansion Assessment

5.1 Overview

The proposed expansion introduces a fully integrated woven sack manufacturing line based on advanced technology. The expansion complements the company's existing recycling, pelletizing, film extrusion, and bag production operations, enabling vertical integration into high-value woven industrial packaging.

The technology selection is anchored on:

- Proven industrial performance
- High energy efficiency
- Stable mass-balance throughput
- Compatibility with recycled and virgin polymer feedstock
- Strong after-sales service presence in the UAE

Objective & Scope of Expansion

- Deploy an **end-to-end woven-sack production cell**: Tape extrusion → Precision winding → Circular looms → Extrusion coating → Flexographic printing → Conversion (cutting & stitching)
- Production capacity limitations, combined with the need to support a broader product variety, prevent the company from effectively serving unaddressed target customer segments.
- Integrate the new capacity with existing recycling/pelletizing and film lines to maximise in-house feedstock conversion and margin capture

5.2 Facility Design & Location Considerations

All of the Company's operations are managed from single location specified at Umm Al Quwain. The existing facility has reached near-full space utilization, with no additional physical capacity available to accommodate new production lines. Any further expansion within the current premises would constrain workflow efficiency, storage capacity, and safety compliance.

Expansion plan

To enable the proposed manufacturing line, the company has identified an adjoining industrial plot of approx. equal size (29,000 sq. ft.). The adjacency ensures seamless integration with current operations while allowing sufficient space for new production infrastructure.

Plant / Location Advantages

- Umm al Quwain industrial zone.
  - Located near major waste generation hubs, the Umm Al Quwain Industrial Zone ensures efficient access to diverse plastic feedstocks
  - Proximity to Sheikh Mohammed Bin Zayed Road and Emirates Road provides rapid connectivity for inbound materials and outbound exports
  - Competitive utility tariffs and free zone incentives (100% ownership, zero corporate tax) lower operating costs
  - Nearby ports (Umm Al Quwain Port and Jebel Ali) facilitate cost-effective distribution across the GCC, South Asia, and East Africa

Proposed layout

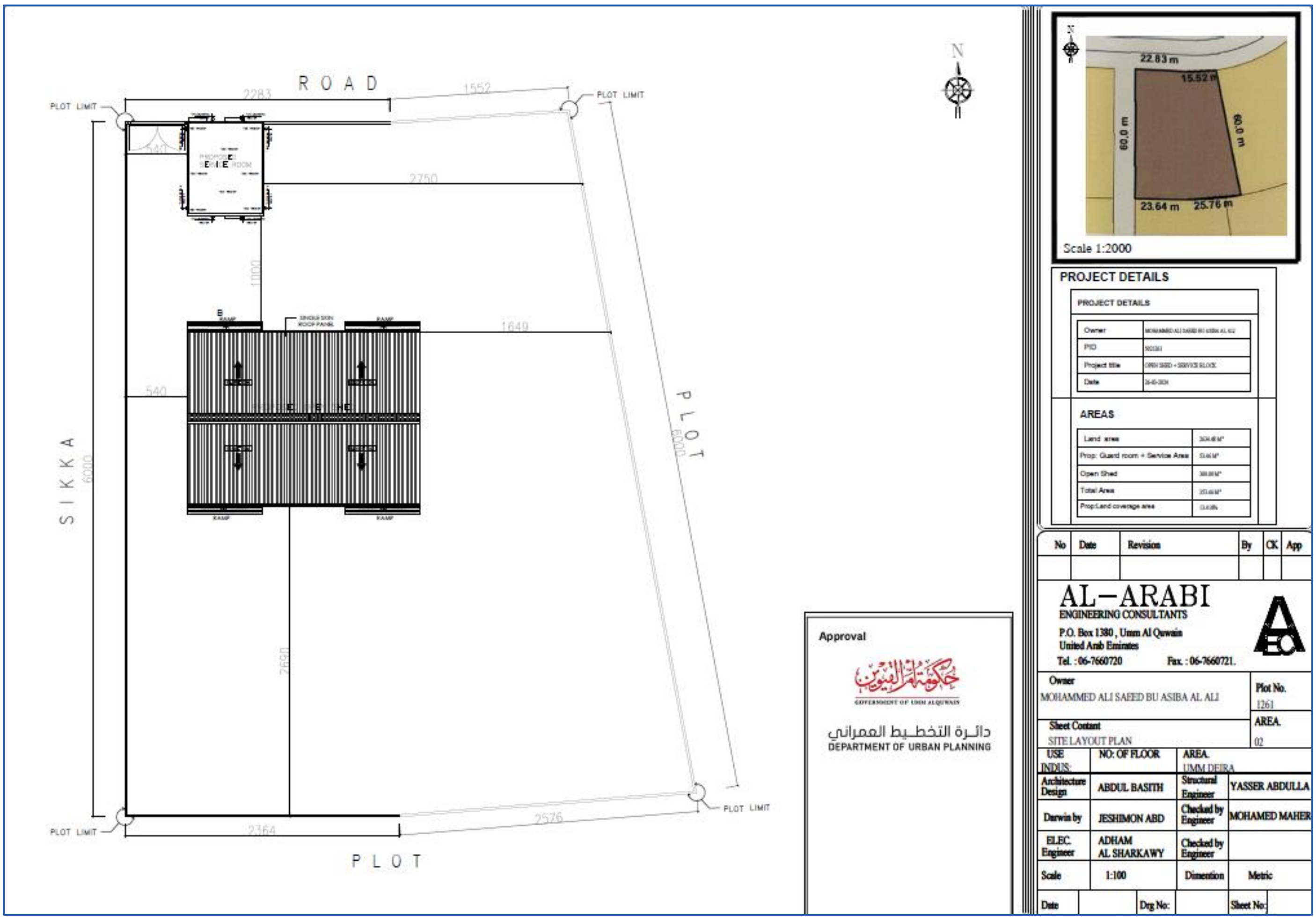


Image 19: Proposed layout

Expansion Plan Details

Land and Building

- Lease of 29,000 sq. ft. adjoining industrial plot
- Functional Sections in Expansion:
  - Tape Line Section
  - Loom Area
  - Unprinted Rolls Area
  - Winder Area
  - Raw Material Storage
  - Finished Goods Storage
  - Office Area (Accounting, Admin)
  - Service Room for Maintenance Teams

Linear process flow (Engineering Sequence):

1. Polymer Preparation & Dosing
2. Tape Extrusion (Lorex - PP/HDPE tape line)
3. In-line Melt Filtration & Melt Metering Pump
4. Stretching / Annealing / Hot-Air Oven (orientation of tapes)
5. Precision Winding (LFW 200CM - tape to bobbins)
6. Bobbins to Circular Loom (Nova 6 - 576 tapes)
7. Woven Fabric Winding (roll build-up with tension control)
8. Extrusion Coating (if required - single/dual side)
9. Flexo Printing (if required - branding & statutory info)
10. Conversion (Cut & Stitch - bag formation, bottom/side sewing)
11. Finished Goods Warehousing & Dispatch

5.3 Technology and Vendor Selection

Overview of Vender Selection Process

The machinery and vendor selection process was conducted through a comprehensive competitive bidding exercise involving

- Lohia Corp Limited
- Wenzhou Kewang Machinery Co. Ltd.
- Exakta Meccanica Ltd.

Each supplier was evaluated on technical specifications, production efficiency, build quality, lifecycle costs, energy consumption, service support, and compatibility with the proposed manufacturing configuration.

The decision was primarily driven by:

- Superior build quality and engineering standards
- Proven operational performance across global woven sack and recycling installations
- Strong service and technical support network within the UAE
- Energy-efficient systems with stable mass-balance throughput
- Lower expected downtime and maintenance risk
- Established brand reputation and decades of industry expertise

Following detailed technical and commercial assessment upon above criteria, Lohia Corp Limited was selected as the exclusive machinery supplier. The decision was primarily driven by its superior build quality, proven industrial performance, and strong after-sales service network within the UAE. Lohia's advanced recycling systems, including the ReclaMax and ReclaPro platforms, are designed to deliver energy-efficient, single-step recycling solutions capable of processing a wide range of plastic materials such as woven sacks and flexible films.

The equipment offers consistent output quality, stable operational performance, and optimized energy utilization. Backed by decades of industry experience and localized technical support, Lohia's machinery provides enhanced reliability and reduced operational risk. The evaluation emphasized technology compatibility with existing operations, production capacity adequacy, energy efficiency, durability, and availability of prompt post-sale service.

While alternative vendors presented competitive proposals, Lohia demonstrated stronger long-term value through superior engineering standards and service assurance. The final selection ensures operational efficiency, reduced downtime risk, and sustainable performance for the expanded recycling and woven packaging production line, thereby strengthening the overall technical and financial viability of the project.

5.4 Proposed Line Components and Utilities

Tape Extrusion Line - Lorex E90B.1000 / Duotec Series

The proposed woven sack production line is anchored by the Lorex E90B.1000 / Duotec tape extrusion system supplied by Lohia Corp Limited. This tape extrusion line serves as the principal mass-balance driver for downstream operations, including winding, weaving, coating, and conversion.

The system is designed specifically for raffia-grade PP/HDPE applications and delivers stable, high-quality tape output suitable for industrial sack production. For the E90B.30D configuration, the melt capacity is ~ 350 kg per hour, which forms the throughput reference for downstream weaving capacity planning.

Key technical features include:

- Barrier screw extruder for improved melt homogeneity
- In-line melt pump ensuring pressure stability and uniform output
- Hot-air ovens for controlled tape stretching
- Combined stretching and annealing units to enhance molecular orientation
- Consistent denier control with minimal variation
- Line speeds of up to ~ 350 meters per minute, depending on tape characteristics

The integrated barrier screw and in-line melt pump configuration ensures uniform melt flow and reduces fluctuations, which is critical for maintaining tape strength and dimensional stability. The advanced stretching and annealing units provide optimal molecular orientation, resulting in high tensile strength and low elongation variation.

Technical/ Design advantages/ Evaluation:

- The Lorex E90B/Duotec line is well suited for a new raffia manufacturing setup
- Barrier screw ensures uniform melting/mixing; melt pump minimizes thickness variation for loom efficiency
- Dual-stage stretching (Duotec) boosts tensile strength, speed, and recycled PP tolerance vs. single-stage rivals
- E90B: 350-400 kg/hr PP, 350-450 m/min; Duotec scales to 900 kg/hr, 600 m/min for future-proofing.

- Anchors mass-balance for downstream weaving/coating with proven reliability
- Lower lifecycle costs, premium products (FIBC/geo-textiles), and ramp-up certainty in TEV-higher CAPEX but superior ROI vs. competitors

Image 20: The Lorex E90B/Duotec line

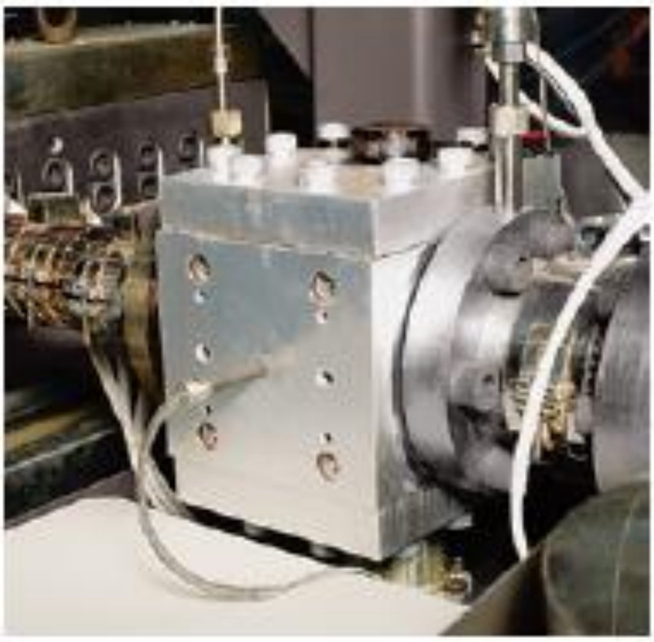


Image 21-22: Melt metering pump, Auto gauging dye

Comparison with Other vendors

Attribute	Lohia Group	Wenzhou Kewang	Exakta Meccanica
Speed/Throughput	350-600 m/min, 900 kg/hr max	200-350 m/min, quality trade-offs	Flexible but lower speeds
Quality/Stability	Low denier variation, advanced controls	Basic for sacks	Niche, less high-volume data
Support	Strong India/Asia network	Logistics risks outside China	Limited footprint

Precision / Filament Winder (LFW 200CM)

The LFW 200CM Precision Filament Winder forms a critical intermediate link between tape extrusion and circular weaving operations. Supplied by Lohia Corp Limited, the winder ensures high-precision bobbin formation, directly influencing loom efficiency and overall fabric quality.

The system is designed for high-speed cross-winding of flat and fibrillated tapes with widths ranging from 1.8 mm to 6 mm and denier ranges between 400 and 3000. Operating speeds between 160 to 450 meters per minute enable the winder to match upstream extrusion output while ensuring smooth feeding into downstream weaving units.

Technical/ Design advantages/ Evaluation:

- Superior bobbin quality, higher weaving efficiency, and local support-optimizes TEV with lower downtime vs. competitors
- 200mm traverse, 35mm ID/218mm bobbin cores, up to 160mm diameter; 6-high/4-across frame (2.1x0.8x1.9m).
- Matches upstream extrusion output, anchors TEV weaving efficiency with stable packages.
- Dancing-arm feedback + individual AC inverter per spindle ensures constant tension for flawless bobbins.
- Adjustable pressure rollers, precise traverse via timing belts/pulleys, and microprocessor sync reduce waste and boost weaving uptime.

Image 23: Precision / Filament Winder (LFW 200CM)

Comparison with Other vendors

Attributes	Lohia LFW 200CM	Wenzhou Kewang	Exakta Meccanica
Tension/Speed	Dancing-arm feedback, 160-450 m/min	Basic tension, 200-350 m/min typical	Flexible but less high-precision data
Package Quality	Excellent uniformity, low breaks	Acceptable for basic sacks	Niche, variable consistency
Controls	Microprocessor + inverters per spindle	Simpler PLC	Custom but less standardized

Circular Loom (Nova 6 - 576 tapes)

Lohia's Nova 6 circular loom (576 tapes) is a six-shuttle, high-speed weaving machine capable of achieving weft insertion speeds of up to 1,100 picks per minute. It is engineered to produce uniform, high-strength tubular woven fabric suitable for applications such as cement, fertilizer, polymer, and animal feed bags, while maintaining optimized energy consumption and low maintenance requirements.

Technical/ Design advantages/ Evaluation:

- For the new plant TEV, Nova 6 is preferred as a slightly higher CAPEX but higher-productivity, lower-risk loom that supports premium bag quality and data-driven efficiency versus typical offerings from Wenzhou Kewang and Exakta Meccanica
- Positive warp in-feed with motor + load-cell feedback maintains precise warp tension, improving fabric GSM consistency and reducing broken ends
- Microprocessor loom controller plus fabric winder with load-cell tension control and LDMS option enable real-time monitoring (breakages, efficiency, downtime) and higher weaving efficiency per loom
- 576 tapes, double-flat working width 30-90 cm; warp/weft bobbins 35 mm ID, 218 mm length; fabric roll diameter up to 1500 mm
- Compact footprint (~9.5 x 2.8 x 3.0 m for 576 tapes) and high insertion rate make it a strong TEV anchor for bag output per loom and overall floor productivity.

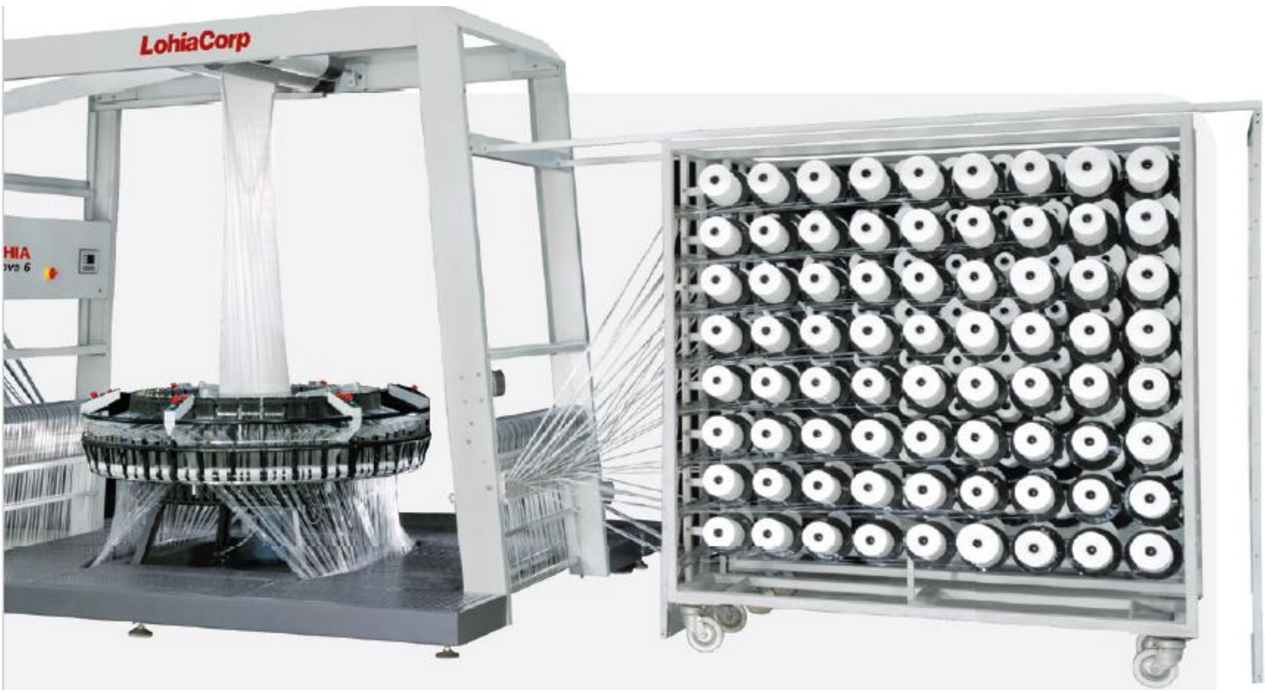


Image 24: Circular Loom

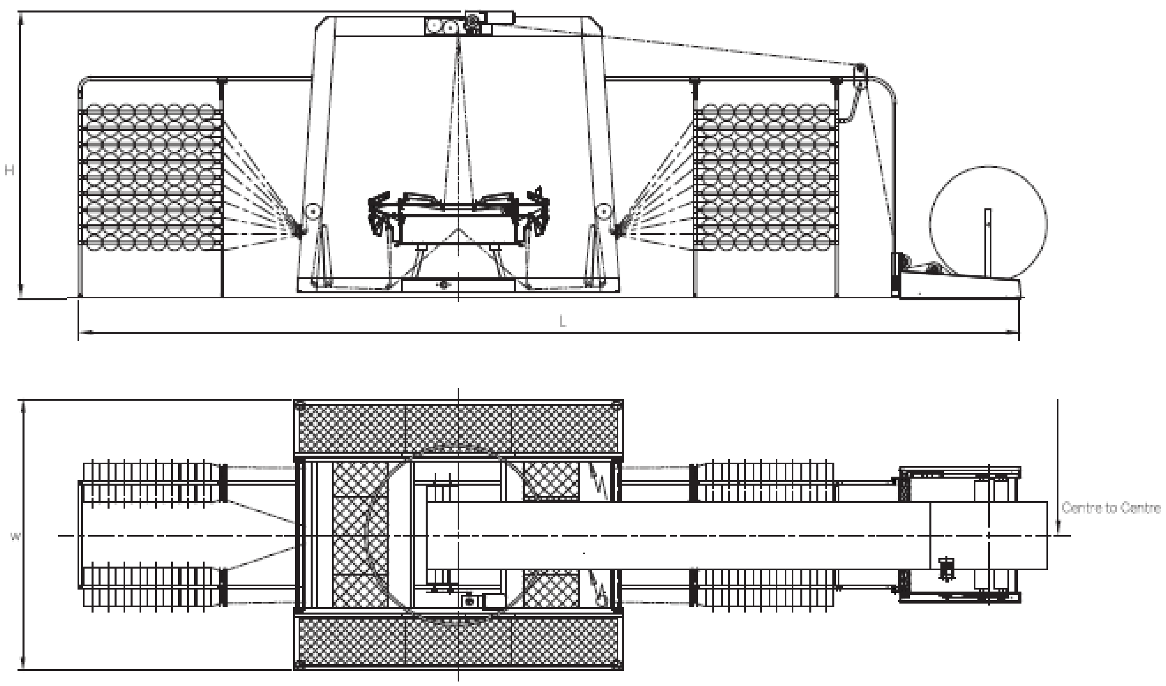


Image 25: Circular loom processing model

Comparison with Other vendors

Attributes	Lohia Nova 6 (576 tapes)	Wenzhou Kewang	Exakta Meccanica
Speed & stability	Up to 1100 picks/min with load-cell warp and fabric tension control for stable high-speed running.	Lower published speeds and simpler tension systems; more de-rating risk in actual operation.	Robust but more conventional controls; less focus on high-speed raffia benchmarks.
Quality & data	Microprocessor controller + optional LDMS for detailed production/stop data and quality control.	Basic counters/PLC; limited integrated data logging.	Custom solutions, but monitoring less standardized.
TEV impact	Higher loom efficiency, fewer breaks, better GSM control and strong India/global support network, lowering lifecycle cost and ramp-up risk.	Lower CAPEX but higher risk on uptime, support, and fabric quality.	Smaller installed base; service and spares less predictable.

Additional equipment (LOHIA)

1. **Extrusion Coating Line** - Inline coating of woven fabric for moisture-barrier, printability and lamination properties
2. **Flexographic Printing Unit** - Multi-pass printing for customer artwork and regulatory information.
3. **Conversion Line (Cutting & Stitching)** - Automated cutting, bottom-stitching, optional gusseting and top-hem; final bale-pack station for dispatch.

Utilities, Site & Erection Considerations

Key utility provisions (from LOHIA):

- **Chiller capacity: ~35 TR** for tape line cooling (specification in LOHIA tape extruder document).
- **Compressed air:** capacity varies with tape-suction guns; LOHIA provides compressor sizing guidance (20 CFM if no guns; up to 165 CFM with two tape guns)

- **Power:** Clean 3-phase supply 415V/50Hz; recommend voltage stabilizer (LOHIA recommended 475 kVA stabilizer for the model in offer)
- **Floor loading & shed heights:** concrete floor with load-bearing capacity ~2.5 MT/m²; minimum shed height recommended (LOHIA advises ~5 m for looms)

LOHIA Group proposes supervised commissioning packages (charges in the offer). Lead-time indicative: **5-6 months** from order confirmation.

Utilities and Labor

Currently, the facility is supported by a 450 kW electricity load, which is sufficient to power both current operations and the planned expansion, including high-capacity extrusion lines, looms, and auxiliary systems. However, it is recommended to get access to 1MW power line for smooth operations.

Water requirements are met through direct connections to Umm Al Quwain's industrial water network, providing a reliable supply for cooling, washing, and sanitation processes. On-site wastewater is routed through the industrial zone's treatment infrastructure, ensuring compliance with environmental regulations and minimizing discharge costs. The Company has approval and clearance from respective utilities providers.

Approvals and Clearances

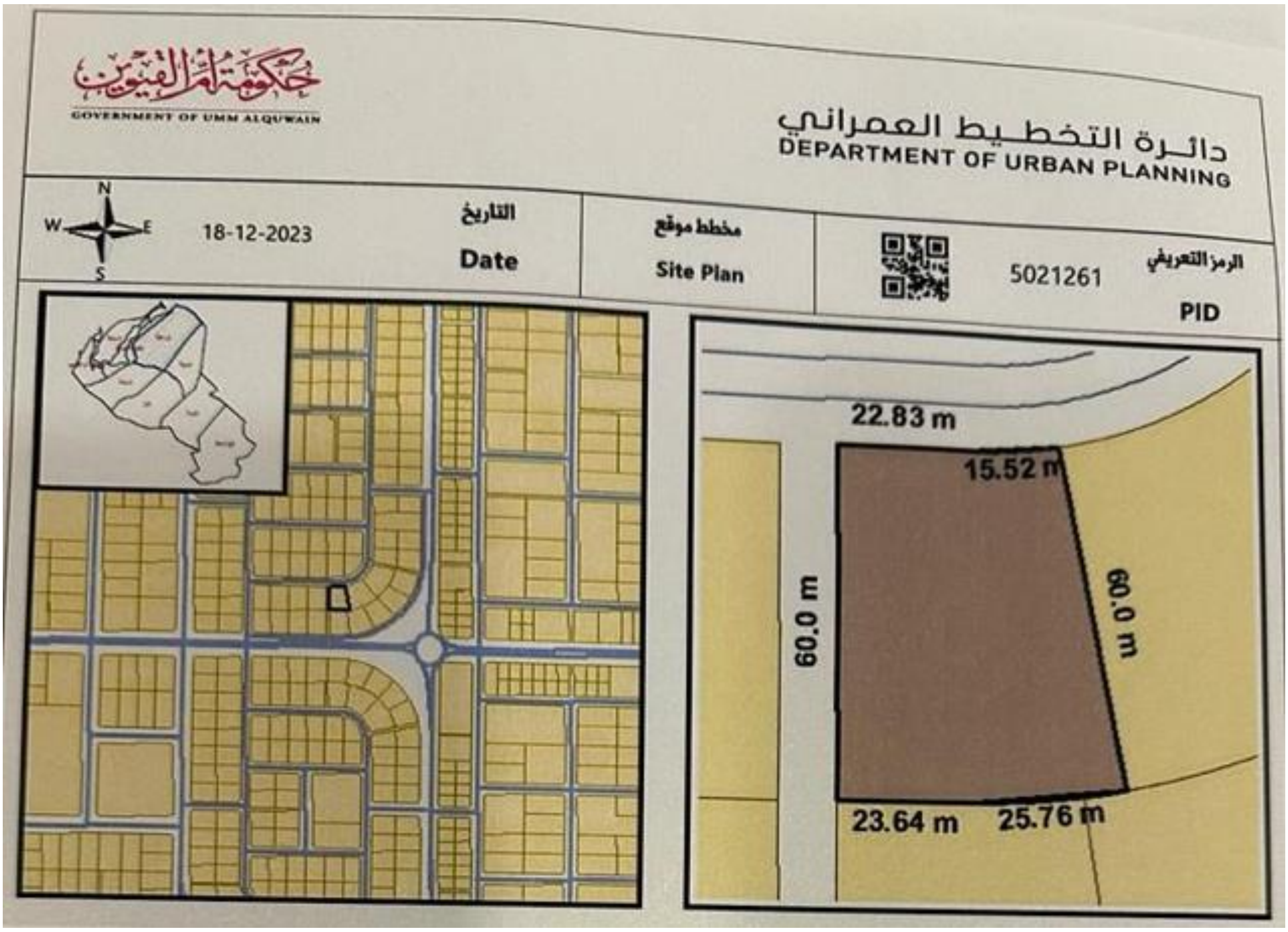


Image 26: Site plan, Department of Urban Planning

الإحداثيات	Lat: 25.55216594	Long: 55.7070174	
إسم المالك Owner Name			
إسم القطاع Sector Name	الزاهية	رقم القطاع Sector No	5
إسم المنطقة Area Name	أم ذرع	رقم المنطقة Area No	02
رقم القطعة Plot No	1261		
المساحة الإجمالية Total Area	2,634.48 m2 / 28,357.54 Ft2	نوع الإستعمال Land Use	صناعي
الارتدادات/ بالمتر Set Back (M)			
نوع الأرض Land Type	بالمنح	عدد الأدوار Total Floors	أرضي+ميزانين+2
شرق East	جار 5	شمال North	شارع 5
رقم الملف File No		رقم الايصال Receipt No	1703-2023-8400 218954
غرب West	سكة 5	جنوب South	جار 5

ملاحظات Notes

1. تم صرف المنحة بتاريخ 09/11/2011
2. في حال عدم تعميرها خلال سنه يلغى الانتفاع تلقائيا
3. يعتبر هذا المخطط بديلا للمخطط السابق الصادر بتاريخ 09/11/2011
4.

المدير العام

رئيس الدائرة

DEPARTMENT OF URBAN PLANNING

Image 27: Approval Department of Urban Planning

Manpower Requirement

Based on the level of automation in technology, it is estimated to have the below mentioned manpower supply (at full capacity) to execution operations.

Category	No. of Persons
Workers	80
Project Manager	1
Sales Head	1
Sales Executives	2
Supervisors	2
Technician	2
Back Office Support Team	2
Driver	4

5.5 Plant Capacity and Throughput Estimation

Prime Design: LOHIA Lorex extruder rated melt capacity ~**350 kg/hr** (PP Raffia grade) - used as the primary mass input for throughput modelling.

Operational assumptions (conservative, TEV-ready):

- Planned **productive hours/day**: 20 (2-shift operation from second year)
- Planned **operating days/month**: 26 (accounts for maintenance & statutory days)
- Planned **operating days/year**: 300 (conservative)

Baseline throughput estimates (material mass):

- **Hourly extrusion melt**: 350 kg/hr.
- **Daily (20 hr)**: 350 × 20 = **7,000 kg/day** (7.0 t/day).
- **Monthly (26 running days)**: 7,000 × 26 = **182,000 kg/month** (182 t).
- **Annual (300 days)**: 7,000 × 300 = **2,100,000 kg/year** (2,100 t).

Downstream conversion & yield assumptions:

- Typical mass losses in tape extrusion → winding → weaving → coating chain (trims, edge waste, moisture): **estimated 6-12%** depending on automation and in-line recycling of trim. With in-line edge-trim recycling (LOHIA edge-trim recycling unit), effective net yield may improve materially.

Loom output (fabric metres/day) - indicative calculation basis:

- Nova 6 looms are specified with high tape counts (576 tapes) and rated weft insertion up to 1,100 picks/min - practical throughput depends on fabric width and GSM. Use manufacturer data to size number of looms to match tape-line daily kg output.

Winder & bobbin match: LFW 200CM winders (160-450 m/min winding range) ensure continuous supply to Looms and reduce downtime due to bobbin changes. Plan winders: loom ratio to avoid starvation at loom creels.

5.6 CAPEX Analysis

Machinery Cost Estimate

- **Project:** 18,000 TPA PP/HDPE Tape Stretching Plant Expansion
- **Equipment Reference:** Technology Plastomech Private Limited and Management discussion on Lohia Group's Machine
- **Model:** tpRex 105/1500 (450 kg/hr per line)
- **Exchange Rate:** 1 USD = 3.67 AED

Estimated Machinery Cost			
	Description	**Price (USD)**	**Price (AED)**
Core Production Machinery			
Section 1	Tape Stretching Line - Model tpRex 105/1500 (Complete with Extruder, Melt Pump, Die, Oven, Godets, Winders, PLC) (5 Sets to be Purchased @ 270,000 USD Per Unit)	1,350,000	4,954,500
Essential Plant Machinery & Utilities			
Section 2	Centralized Raw Material Handling System	120,000	440,400
	Air Compressor System	60,000	220,200
	Cooling Tower & Chilling System	150,000	550,500
	DG Backup (750-1000 kVA)	200,000	734,000
	Electrical Panels, Transformers & Cabling	180,000	660,600
	Material Handling Equipment (Forklifts etc.)	70,000	256,900
	Quality Control & Lab Equipment	40,000	146,800
	Installation, Erection & Commissioning	1,00,000	367,000
	Sub Total (Utilities & Supporting Machinery)	**920,000**	**3,376,400**
	Total Cost of Machinery	**2,270,000**	**8,330,900**
	Provision of Price fluctuation (@5%)	113,500	416,545
	Total Estimated Cost of Machinery	**2,383,500**	**8,747,445**

Estimated Civil and Infrastructure Cost	
Work Covered	**Indicative Cost (AED)**
Site Preparation & Earthworks	104,000
Structure & Foundations	390,000
Roof & Insulation	104,000
Walls / Cladding	130,000
MEP & Utilities	156,000
Doors, Access & Loading	65,000
Office Finishes & Amenities	65,000
External Works	65,000
Contingency	52,000
Total Cost of Construction	**1,131,000**
Provision of Price fluctuation (@5%)	56,550
Total Estimated Cost of Construction	**1,187,550**

Capex Breakdown	Amount (AED)
Machinery Cost	8,747,445
Civil and Infrastructure Setup Cost	1,187,550
Working Capital	8,465,005
Total Capex	**18,400,000**

Financial Analysis

This section provides a comprehensive financial assessment of the company's projected performance from 2025 to 2035, with particular focus on evaluating the economic feasibility of the proposed capacity expansion from 6,000 MT to 24,000 MT. The analysis is based on the unaudited FY2025 financials (as provided by management) and forward projections incorporating production ramp-up, pricing assumptions, cost behaviour, and financing structure.

The financial evaluation aims to assess:

- Revenue scalability and absorption of expanded capacity
- Margin sustainability under rising input costs
- EBITDA stability and operating leverage
- Debt servicing capability for AED 14.4 million external borrowing and AED 4 million of Promoter's debt
- Return on investment and payback profile
- Overall long-term financial viability of the project

The objective is to determine whether the proposed expansion generates adequate cash flows, supports repayment obligations, and enhances shareholder value while maintaining acceptable risk levels during the ramp-up phase.

6.1 Overview of Financial Performance (FY2025)

The financial performance for FY2025 (unaudited, as shared by management) reflects operations based solely on the existing installed capacity of 6,000 MT. During the year, the company operated at ~ 82% utilization, producing 4,920 MT and generating total revenue of AED 11.3 million.

The average selling price stood at AED 2,300 per ton, while the raw material purchase price averaged AED 1,650 per ton, indicating a gross spread of AED 650 per ton before conversion and operating costs. Total cost of goods sold amounted to AED 9.31 million, resulting in a gross profit of AED 2.0 million, translating to a gross margin of ~ 17.7%.

After accounting for indirect and administrative expenses, EBITDA for 2025 was AED 0.50 million, reflecting a modest EBITDA margin of ~ 4.4%. The relatively low margin profile is attributable to limited economies of scale under the current capacity structure and relatively fixed overhead absorption across a smaller production base.

The gross margin for the year stood at ~ 17-18%, reflecting a healthy spread between selling price and raw material cost. However, due to limited scale and relatively fixed overhead structure, EBITDA margin remained modest at ~ 4-5%.

Depreciation expenses of AED 1.60 million resulted in a reported loss before tax of AED 1.09 million. Despite the accounting loss, the company generated positive operating cash flow, indicating that the core business operations are cash-generative but constrained by limited production scale.

The 2025 financial profile highlights the following structural observations:

- Adequate gross spreads at current pricing levels
- Limited margin expansion due to fixed cost absorption constraints
- Positive operating cash flow despite accounting losses
- Capacity constraints restricting revenue growth potential

Overall, FY2025 reflects a stable but scale-limited operation. The financial performance demonstrates that while the existing business model is operationally viable, significant margin enhancement and sustainable profitability require higher production volumes and improved fixed cost absorption-both of which are addressed through the proposed capacity expansion.

Strategic Rationale for Capacity Expansion - Financial Perspective

The financial performance in FY2025 clearly demonstrates that while the company's core operations are commercially viable and cash-generative at the gross level, profitability remains constrained by limited production scale and sub-optimal absorption of fixed costs. With installed capacity capped at 6,000 MT and utilization already at 82%, the existing facility has limited headroom for revenue growth without incremental capital investment.

Key financial limitations observed in FY2025 include:

- Modest EBITDA margin due to fixed overhead absorption across a smaller production base
- Inability to materially improve profitability without scale enhancement
- Capacity ceiling restricting revenue expansion despite healthy market demand
- Limited operational leverage under current infrastructure

The proposed expansion from 6,000 MT to 24,000 MT is therefore not merely growth-oriented but structurally necessary to unlock operating leverage. Higher production volumes are expected to:

- Improve fixed cost absorption
- Enhance EBITDA margins
- Strengthen cash flow generation
- Improve debt servicing capability
- Deliver sustainable long-term profitability

From a financial standpoint, the expansion transforms the business from a small-scale processing unit into a scaled industrial packaging and recycling operation capable of generating stable returns and supporting structured debt financing. The shift in capacity is expected to materially improve economic efficiency and overall financial resilience.

6.2 Projected Financial Post Capacity Expansion

Particulars	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
Production (MT)	4,920	9,300	12,300	15,150	17,250	19,350	20,400	20,400	20,400	20,400	20,400
Capacity Utilization (%)	82%	53%	55%	58%	63%	68%	70%	70%	70%	70%	70%
Sales Revenue	11.32	22.46	31.19	40.34	48.23	56.80	62.88	66.02	69.32	72.79	76.43
Raw Material Cost	(8.12)	(16.11)	(22.38)	(28.94)	(34.60)	(40.75)	(45.11)	(47.36)	(49.73)	(52.22)	(54.83)
Direct Expenses	(1.20)	(3.46)	(4.23)	(4.97)	(5.67)	(6.43)	(6.94)	(7.19)	(7.46)	(7.74)	(8.03)
COGS	(9.31)	(19.57)	(26.61)	(33.91)	(40.27)	(47.17)	(52.05)	(54.56)	(57.19)	(59.95)	(62.86)
Gross Profit	**2.00**	**2.89**	**4.58**	**6.43**	**7.96**	**9.63**	**10.83**	**11.47**	**12.13**	**12.84**	**13.57**
Indirect Expenses	(1.50)	(2.75)	(3.30)	(3.73)	(4.11)	(4.53)	(5.05)	(5.30)	(5.57)	(5.85)	(6.15)
EBITDA	0.50	0.14	1.28	2.71	3.84	5.09	5.78	6.16	6.56	6.99	7.43
EBITDA Margin (%)	4.4%	0.6%	4.1%	6.7%	8.0%	9.0%	9.2%	9.3%	9.5%	9.6%	9.7%
Depreciation	(1.60)	(2.24)	(2.24)	(2.24)	(2.24)	(2.24)	(2.24)	(2.24)	(2.24)	(2.24)	(2.24)
Interest Cost	-	(2.31)	(1.50)	(1.33)	(1.15)	(0.95)	(0.73)	(0.50)	(0.36)	(0.36)	(0.36)
Profit Before Tax (PBT)	**(1.10)**	**(4.41)**	**(2.46)**	**(0.87)**	**0.45**	**1.90**	**2.81**	**3.43**	**3.96**	**4.38**	**4.83**
Profit After Tax (PAT)	(1.10)	(4.41)	(2.46)	(0.87)	0.07	1.39	2.21	2.78	3.27	3.65	4.05
Operating Cash Flow	**0.50**	**(2.17)**	**(0.22)**	**1.38**	**2.31**	**3.63**	**4.45**	**5.02**	**5.51**	**5.89**	**6.29**

The projected financial performance post capacity expansion reflects a strong scale-up in operations, driven by the addition of 18,000 MT of new capacity and improved product mix towards higher-value industrial packaging. The expansion enables the company to transition from a relatively small-scale recycling and bag manufacturing unit to an integrated, higher-margin packaging player. While the initial years (2026-2028) reflect a ramp-up phase with moderate financial pressure, the business demonstrates a clear trajectory toward stable profitability, improved margins, and strong cash flow generation in the steady-state period.

Revenue is expected to grow significantly, supported by both volume expansion and moderate price escalation. At the same time, operating leverage plays a critical role in improving profitability as fixed costs are absorbed over higher production volumes. The projections indicate that the business achieves operational stabilization by Year 3-4, followed by consistent growth and strong financial performance thereafter.

(AED, Million)

Particulars	2025 Actual Unaudited (As shared)	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
Old machine Capacity	6,000	6,000	6,000	6,000	6,000	6,000	6,000	6,000	6,000	6,000	6,000
New machine Capacity	-	18,000	18,000	18,000	18,000	18,000	18,000	18,000	18,000	18,000	18,000
Installed Capacity (MT)	6,000	24,000	24,000	24,000	24,000	24,000	24,000	24,000	24,000	24,000	24,000
Old machine Production	4,920	4,800	3,300	3,450	3,750	4,050	4,200	4,200	4,200	4,200	4,200
New machine Production	-	4,500	9,000	11,700	13,500	15,300	16,200	16,200	16,200	16,200	16,200
Production (MT)	4,920	9,300	12,300	15,150	17,250	19,350	20,400	20,400	20,400	20,400	20,400
Old machine Utilization	82%	80%	60%	50%	50%	50%	50%	50%	50%	50%	50%
New machine Utilization	0%	25%	50%	65%	75%	85%	90%	90%	90%	90%	90%
Capacity Utilisation (%)	82%	53%	55%	58%	63%	68%	70%	70%	70%	70%	70%
Sales Price	2,300.00	2,415.00	2,535.75	2,662.54	2,795.66	2,935.45	3,082.22	3,236.33	3,398.15	3,568.05	3,746.46
Raw Material Purchase Price	1,650.00	1,732.50	1,819.13	1,910.08	2,005.59	2,105.86	2,211.16	2,321.72	2,437.80	2,559.69	2,687.68
REVENUE											
Sales Revenue	11.32	22.46	31.19	40.34	48.23	56.80	62.88	66.02	69.32	72.79	76.43
Total Revenue (A)	11.32	22.46	31.19	40.34	48.23	56.80	62.88	66.02	69.32	72.79	76.43
COST OF RAW MATERIALS											
Raw Material Purchase	8.12	16.11	22.38	28.94	34.60	40.75	45.11	47.36	49.73	52.22	54.83
Total Raw Material Cost (B)	8.12	16.11	22.38	28.94	34.60	40.75	45.11	47.36	49.73	52.22	54.83
Direct Manufacturing Expenses											
Diesel	0.01	0.02	0.03	0.04	0.04	0.05	0.06	0.06	0.06	0.07	0.07
Factory Expenses	0.18	0.38	0.55	0.71	0.85	1.01	1.11	1.17	1.23	1.29	1.35
Factory Kitchen	0.00	0.01	0.01	0.01	0.02	0.02	0.02	0.02	0.03	0.03	0.03
Factory Van	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.01	0.01	0.01
Factory FEWA (Power)	0.33	1.31	1.37	1.44	1.56	1.68	1.75	1.75	1.75	1.75	1.75
Factory FEWA (Water)	-	0.11	0.15	0.18	0.21	0.23	0.24	0.25	0.25	0.26	0.26
Labour – Cutting	0.01	0.02	0.02	0.03	0.04	0.05	0.05	0.05	0.06	0.06	0.06
Chemicals	0.00	0.01	0.01	0.01	0.02	0.02	0.02	0.02	0.02	0.03	0.03
Packing Material	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.07	0.07	0.08	0.08
Purchase Commission	0.00	0.01	0.01	0.01	0.01	0.01	0.02	0.02	0.02	0.02	0.02
Repairs & Maintenance	0.02	0.05	0.07	0.09	0.11	0.13	0.14	0.15	0.16	0.17	0.17
Factory Rent	0.31	0.62	0.65	0.68	0.72	0.75	0.79	0.83	0.87	0.92	0.96
Factory Wages	0.31	0.91	1.32	1.71	2.04	2.40	2.66	2.79	2.93	3.08	3.23
Total Direct Expenses	1.20	3.46	4.23	4.97	5.67	6.43	6.94	7.19	7.46	7.74	8.03
COST OF GOODS SOLD	9.31	19.57	26.61	33.91	40.27	47.17	52.05	54.56	57.19	59.95	62.86
GROSS PROFIT (A – COGS)	2.00	2.89	4.58	6.43	7.96	9.63	10.83	11.47	12.13	12.84	13.57
Indirect / Administrative Expenses											
Sales and Marketing	-	0.10	0.10	0.11	0.12	0.13	0.15	0.16	0.18	0.19	0.21
Office Rent	0.09	0.17	0.26	0.27	0.28	0.30	0.31	0.33	0.34	0.36	0.38
Accounting & Audit	0.01	0.01	0.01	0.01	0.01	0.02	0.02	0.02	0.02	0.02	0.02
Bank Charges	0.05	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.07	0.07	0.07
Vehicle & Transport	0.16	0.34	0.49	0.64	0.76	0.90	1.00	1.05	1.10	1.15	1.21

Particulars	2025 Actual Unaudited (As shared)	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
Salaries	0.88	1.32	1.45	1.52	1.60	1.68	1.94	2.04	2.14	2.25	2.36
Staff Welfare	-	0.12	0.12	0.12	0.13	0.13	0.13	0.14	0.14	0.14	0.14
Licenses & Legal	0.19	0.21	0.22	0.23	0.25	0.26	0.27	0.29	0.30	0.31	0.33
Logistics & Travel	0.03	0.07	0.10	0.13	0.16	0.18	0.20	0.21	0.22	0.23	0.25
Other Admin	0.17	0.35	0.48	0.62	0.74	0.88	0.97	1.02	1.07	1.12	1.18
Total Indirect Expenses (D)	1.59	2.75	3.30	3.73	4.11	4.53	5.05	5.30	5.57	5.85	6.15
Total Cost	10.90	22.32	29.91	37.63	44.38	51.71	57.10	59.86	62.76	65.80	69.00
EBITDA (Gross Profit – Indirect Expenses)	0.42	0.14	1.28	2.71	3.84	5.09	5.78	6.16	6.56	6.99	7.43
EBITDA Margin	3.69%	0.63%	4.11%	6.71%	7.97%	8.97%	9.19%	9.34%	9.47%	9.60%	9.72%
Depreciation (10%)	1.60	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24
Interest on Outside Debt	-	1.95	1.14	0.97	0.79	0.59	0.37	0.14	-	-	-
Interest on Promoter's Debt	-	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.36
PROFIT BEFORE TAX (PBT)	-1.18	-4.41	-2.46	-0.87	0.45	1.90	2.81	3.43	3.96	4.38	4.83
Exemption in Income Tax	-	-	-	-	0.38	0.38	0.38	0.38	0.38	0.38	0.38
Profit Taxable	-1.10	-4.41	-2.46	-0.87	0.08	1.53	2.43	3.05	3.59	4.01	4.45
Tax @ 9%	-	-	-	-	0.01	0.14	0.22	0.27	0.32	0.36	0.40
PROFIT AFTER TAX (PAT)	-1.10	-4.41	-2.46	-0.87	0.07	1.39	2.21	2.78	3.27	3.65	4.05
Operating Cash Flow (PAT+Decpreciation)	0.50	-2.17	-0.22	1.38	2.31	3.63	4.45	5.02	5.51	5.89	6.29

Revenue Growth and Capacity Utilization

Cost structure is predominantly variable in nature, with raw material forming the largest cost component, while margins improve progressively with scale and operational efficiency.

- Raw material cost accounts for ~70-75% of total revenue, making it the key cost driver
- Direct manufacturing expenses constitute ~10-12% of revenue and scale with production volumes
- Total variable cost remains at ~80-85% of revenue in steady state
- Indirect / administrative expenses remain relatively stable and decline as a percentage of revenue with scale
- EBITDA improves from AED 0.14 million (2026) to AED 7.43 million (2035)
- Margin improvement is driven by:
  - Better capacity utilization and fixed cost absorption
  - Economies of scale in direct manufacturing costs
  - Improved product mix with higher-value industrial packaging
- Operating leverage becomes visible post Year 3, with significant margin expansion thereafter
- Cost structure remains sensitive to:
  - Raw material price fluctuations
  - Selling price variations (spread management critical)

The projections indicate a scalable cost structure with improving margins, supported by efficient cost management and realistic ramp-up assumptions, ensuring sustainable profitability in the steady state.

Cost Structure and Margins

Cost structure is predominantly variable, with raw material forming the largest component, while margins improve steadily with scale, better capacity utilization, and operating efficiencies.

- Raw material cost constitutes ~70-75% of total revenue, making it the primary cost driver
- Direct manufacturing expenses account for ~10-12% of revenue and increase proportionately with production
- Total variable cost remains at ~80-85% of revenue in steady state
- Indirect / administrative expenses remain relatively stable and decline as a percentage of revenue with scale
- EBITDA improves from AED 0.14 million (2026) to AED 7.43 million (2035)
- EBITDA margins expand from ~0.6% (2026) to ~9-10% in steady state
- Contribution per ton stabilizes at ~AED 500+ per ton, supporting fixed cost absorption
- Margin expansion is supported by:
  - Improved capacity utilization and operating leverage
  - Economies of scale in direct costs beyond ~50% utilization
  - Better product mix with higher-value woven and industrial packaging
- Fixed cost base (admin + depreciation) remains controlled, enabling strong operating leverage post ramp-up
- Cost structure remains sensitive to:
  - Raw material price volatility
  - Selling price fluctuations (spread between price and RM is critical)

The projections reflect a scalable and efficient cost structure, with gradual margin expansion and sustainable profitability as operations stabilize and capacity utilization improves.

Detailed Financial Feasibility Evaluation

Cash Flow Used for Evaluation

For project evaluation, operating cash flows from 2026-2035 have been considered, as 2025 reflects pre-expansion operations. The expansion investment (external debt + promoter loan) is treated as initial capital outflow.

- Initial Investment
- External Debt: 14.4 (AED, Million)
- Promoter Loan: 4 (AED, Million)
- Total Investment Considered: 18.4 (AED, Million)

(AED,Million)

Year	EBITDA	Operating Cash Flow
2025	0.50	0.50
2026	0.14	(2.17)
2027	1.28	(0.22)
2028	2.71	1.38
2029	3.84	2.31
2030	5.09	3.63
2031	5.78	4.45
2032	6.16	5.02
2033	6.56	5.51
2034	6.99	5.89
2035	7.43	6.29

The financial evaluation of the proposed capacity expansion is based on projected operating cash flows over the period 2026 to 2035, reflecting the project's ability to generate sufficient internal accruals to support debt servicing, capital recovery, and long-term sustainability. The cash flow profile follows a typical manufacturing investment cycle, with initial pressure during the ramp-up phase, followed by steady improvement as capacity utilization increases and margins stabilize. The projections demonstrate that the business transitions into a strong cash-generating entity from the stabilization phase onward, supporting overall financial viability.

- The initial negative cash flow phase is temporary and consistent with project ramp-up dynamics
- The project demonstrates strong cash flow generation capacity in steady state, supporting long-term sustainability
- Increasing cash flows contribute to improved financial flexibility and reduced reliance on external funding
- The overall cash flow trajectory reinforces the project's ability to meet financial obligations and generate returns
- Operating cash flows are negative in the initial years (2026-2027) due to:
    - Low capacity utilization during ramp-up
    - High fixed cost absorption
    - Interest burden from project financing
- Cash flows turn positive from 2028 onward, indicating operational stabilization
- Operating cash flow improves from AED (2.17) million (2026) to AED 6.29 million (2035)
- Cumulative cash flows indicate recovery of initial investment within ~7 years, reflecting a reasonable payback period for a capital-intensive project
- From 2029 onward, the business generates consistent and stable cash flows, enabling:
    - Comfortable debt servicing
    - Funding of working capital requirements
    - Potential for reinvestment and expansion
- Peak operating cash flows exceed AED 6 million annually in steady state, indicating strong internal accrual generation
- The cash flow profile supports:
    - DSCR improvement beyond 1.5x in steady state
    - Sustainable financial structure with declining leverage risk over time

Project IRR (Unlevered IRR)

The Project IRR (Unlevered IRR) for the proposed expansion is estimated in the range of **~18%–20%**, reflecting the intrinsic return generated by the project's operating cash flows independent of its financing structure. This level of return is considered strong for a capital-intensive manufacturing investment in the plastic recycling and packaging sector, particularly given the phased ramp-up and conservative utilization assumptions embedded in the projections.

The return profile is moderately back-ended, with lower returns observed during the initial years (2026–2028) due to underutilization, fixed cost absorption, and delayed cash flow generation. However, from the stabilization phase onward, improved capacity utilization, operating leverage, and margin expansion drive a significant increase in cash flows, resulting in a robust and sustainable return profile over the project life.

Key Observations

- The project IRR exceeds typical industry hurdle rates, indicating strong economic viability and value creation potential

- The return profile is back-loaded, with a substantial portion of value generated during the steady-state operating phase
- The project demonstrates resilience to moderate downside scenarios, though returns remain sensitive to margin compression factors such as pricing and raw material volatility
- The presence of operating leverage significantly enhances returns as utilization stabilizes, reinforcing scalability of the business model

Payback Period

The project is expected to achieve payback within ~7 years, with cumulative cash flows turning positive around FY 2033. The payback trajectory reflects a structured recovery pattern, where initial cash flow deficits during the ramp-up phase are gradually offset by increasing operating cash inflows as production levels and margins improve.

The recovery profile is back-ended, with a significant portion of capital recovery occurring during the stabilization and steady-state phases. While the initial years are impacted by lower utilization and financing costs, the strong improvement in EBITDA and operating cash flows from FY 2029 onward accelerates recovery, ensuring that the investment is recouped within an acceptable timeframe for such projects.

Key Observations:

- The payback period is within acceptable benchmarks for industrial manufacturing projects, supporting investment attractiveness
- The recovery profile is dependent on achieving targeted utilization levels, with delayed ramp-up directly impacting the recovery timeline
- Strong operating leverage and improving margins play a critical role in accelerating payback during later years
- The project maintains the ability to recover investment even under moderate stress scenarios, though with extended timelines

(AED,Million)

Year	Operating Cash Flow	Cumulative Cash Flow	Remaining Amount to Recover
Initial Investment	-	-	(18.40)
2026	(2.17)	(2.17)	(20.57)
2027	(0.22)	(2.39)	(20.79)
2028	1.38	(1.01)	(19.41)
2029	2.31	1.30	(17.10)
2030	3.63	4.93	(13.47)
2031	4.45	9.38	(9.02)
2032	5.02	14.40	(4.00)
2033	**5.51**	**19.91**	**1.51 (Pay back)**
2034	5.89	25.80	-
2035	6.29	32.09	-

The detailed payback analysis confirms that the proposed expansion achieves capital recovery within a reasonable timeframe, supported by a clear transition from initial ramp-up stress to strong and sustained cash generation. The back-ended yet predictable recovery profile, combined with improving operational efficiency and margin expansion, reinforces the financial viability and bankability of the project.

DSCR (Debt Service Coverage Ratio)

The DSCR profile reflects the project's ability to service its debt obligations through operating cash flows, with a clear transition from initial stress to strong repayment capacity. In the early years (2026–2027), DSCR remains below 1.0x due to ramp-up constraints, lower EBITDA, and higher interest burden, which is typical for expansion-stage manufacturing projects.

From FY 2028 onward, DSCR improves steadily as operations stabilize, crossing ~1.5x by FY 2029 and reaching ~1.8x–2.5x in steady state. This improvement is driven by enhanced EBITDA generation, better cost absorption, and a declining interest burden over time, indicating a strengthening financial position and reduced credit risk.

(AED,Million)

Year	EBITDA	Estimated Debt Service	DSCR
2026	0.14	3.80	0.04x
2027	1.28	3.00	0.43x
2028	2.71	2.80	0.97x
2029	3.84	2.60	1.48x
2030	5.09	2.40	2.12x
2031	5.78	2.20	2.63x
2032	6.16	2.00	3.08x
2033	6.56	1.50	4.37x
2034	6.99	1.00	6.99x
2035	7.43	0.80	9.29x

Key Observations

- The initial DSCR weakness is temporary and aligns with expected ramp-up dynamics rather than structural financial issues
- The project demonstrates strong improvement in debt servicing capability post stabilization, providing comfort to lenders
- Steady-state DSCR levels exceed standard lending thresholds, indicating adequate cushion for debt repayment
- The DSCR profile remains sensitive to margin fluctuations, particularly under combined stress scenarios, though not critically impaired
- Strong improvement post Year 3 driven by:
    - Increase in production volumes
    - Better fixed cost absorption
    - Margin stabilization
- DSCR remains **above lender comfort levels (≥1.5x)** in steady state, indicating:
    - Adequate debt servicing capability
    - Lower default risk
- Declining interest burden over time further enhances DSCR in later years
- Cash flow stability ensures consistent coverage even under moderate stress scenarios

Break-Even Analysis

The estimated break-even utilization level for the expanded facility is approximately ~55%–60% of installed capacity, based on steady-state cost and revenue assumptions. This reflects the level at which contribution margins are sufficient to cover fixed costs, including administrative overheads and depreciation.

The financial projections assume a steady-state utilization of around ~70%, providing a safety margin of approximately 10–15 percentage points above break-even. This operational buffer ensures that the business can withstand moderate fluctuations in demand, pricing, or input costs without materially impacting its profitability or financial stability.

Particulars	Value	Per Ton (AED)
Installed Capacity (MT)	24,000	-
Actual Production (MT)	20,400	-
Sales Revenue (AED,Million)	62.88	3,082
Raw Material Cost (AED,Million)	(45.11)	(2,211)
Direct Expenses (AED,Million)	(6.94)	(340)
Total Variable Cost (AED,Million)	(52.05)	(2,551)
Contribution (AED,Million)	10.83	531
Indirect Expenses (Fixed) (AED,Million)	(5.05)	-
Depreciation (Fixed) (AED,Million)	(2.24)	-
Total Fixed Cost (AED,Million)	(7.29)	-
EBITDA (Check) (AED,Million)	5.78	-

Contribution per ton ≈ AED 531

Metric	Value
Break-Even Production (MT)	~13,700 MT
Installed Capacity (MT)	24,000 MT
Break-Even Utilization (%)	~57%
Steady-State Utilization	~70%
Utilization Cushion	~13%
Break-Even Revenue	~AED 42.3 Mn

Key Observations

- The break-even level is achievable within the early stabilization phase, indicating a manageable operational risk profile
- The presence of a utilization buffer enhances resilience against demand variability and market uncertainties
- Contribution margins are adequate to support fixed cost absorption at moderate utilization levels
- Break-even sensitivity to pricing and raw material costs highlights the importance of effective margin management

Profitability Trend

The projected profitability trend reflects a gradual transition from initial losses during the ramp-up phase to stable and improving earnings in the steady-state period. In the early years (2026–2028), profitability is impacted by low capacity utilization, higher fixed cost absorption, and interest burden, resulting in subdued EBITDA margins and negative net profits. However, as production scales up and utilization improves, the business benefits from operating leverage, leading to a steady expansion in margins and profitability.

From FY 2029 onward, the company achieves consistent profitability, supported by higher production volumes, improved cost absorption, and a stable contribution margin. EBITDA continues to grow in line with revenue, while net profitability strengthens as interest costs decline over time. By the steady-state period, the business demonstrates a stable margin profile and sustainable earnings, indicating a mature and efficient operating structure.

Key Observations

- Profitability follows a typical ramp-up curve, with initial losses transitioning into stable profits as operations scale up
- EBITDA shows consistent growth from AED 0.14 million (2026) to AED 7.43 million (2035), reflecting strong operating performance
- EBITDA margins improve from ~0.6% in initial years to ~9–10% in steady state, driven by operating leverage and cost efficiency
- Net profitability turns positive from FY 2029 onward, indicating successful absorption of fixed costs and financing expenses
- Margin expansion is primarily supported by:
  - Improved capacity utilization
  - Economies of scale in manufacturing
  - Better product mix in woven and industrial packaging
- Declining interest burden over time contributes to improved net profit margins in later years
- Profitability remains sensitive to spread (selling price vs raw material cost), making margin management critical
- The business demonstrates stable and predictable earnings post stabilization, enhancing financial visibility and lender confidence

The profitability trend indicates a well-structured growth trajectory, with temporary pressure during the initial phase followed by sustained margin expansion and stable earnings. The ability to achieve consistent profitability within a reasonable timeframe, supported by operating leverage and efficient cost management, reinforces the financial viability and long-term sustainability of the proposed capacity expansion.

6.3 Sensitivity Analysis

Sensitivity analysis indicates that the project remains financially viable under moderate stress scenarios. A 5% decline in selling price or a 5% increase in raw material cost reduces margins and IRR but still keeps returns above acceptable industrial benchmarks. Similarly, a one-year delay in ramp-up extends payback but does not materially impair long-term viability. These results demonstrate that the project has reasonable resilience to pricing and cost fluctuations.

Base Case:

(AED,Million)

Particulars	Base Case	Strategic Comments
Sales Revenue	62.88	Represents steady-state operations with normalized utilization and pricing.
Raw Material Cost	(45.11)	Stable procurement and cost structure assumed.
Direct Expenses	(6.94)	Reflects efficient scale-driven cost absorption.
Contribution	10.83	Healthy contribution margin supporting profitability.
Indirect Expenses	(5.05)	Fixed overheads well absorbed at steady-state scale.
EBITDA	5.78	Strong operating profitability.
EBITDA Margin (%)	9.2%	Stable margin profile.
Contribution / Ton (AED)	531	Efficient unit economics.
Break-Even Utilization (%)	57%	Comfortable cushion vs actual utilization.
DSCR (Steady-State)	~1.8x-2.2x	Strong debt servicing capability.
Project IRR	~18-20%	Attractive returns.
Payback Period	~7 years	Acceptable recovery period.

Scenario 1: 5% Reduction in Selling Price

Under this scenario, the average selling price declines by 5% while production volumes and raw material costs remain unchanged. The reduction directly compresses contribution margins, significantly lowering EBITDA due to the high operating leverage nature of the business. While the project remains operationally viable, margin compression materially weakens profitability and increases break-even utilization levels. The long-term IRR declines but remains above minimum acceptable industrial thresholds.

(AED,Million)

Particulars	Base Case	-5% Price	Change	Strategic Comments
Sales Revenue	62.88	59.73	(3.15)	Revenue directly impacted by pricing pressure
Raw Material Cost	(45.11)	(45.11)	-	No change; highlights pricing sensitivity
Direct Expenses	(6.94)	(6.94)	-	Fixed/semi-variable; limited flexibility
Contribution	10.83	7.68	(3.15)	Significant drop due to reduced price spread
Indirect Expenses	(5.05)	(5.05)	-	Fixed overheads compress margins further
EBITDA	5.78	2.64	(3.14)	Sharp decline (~54%)
EBITDA Margin (%)	9.2%	4.4%	↓	Margin nearly halves
Contribution / Ton (AED)	531	376	(155)	Weakening unit economics
Break-Even Utilization (%)	57%	~72-75%	↑	Reduced operating cushion

DSCR (Steady-State)	~1.8x-2.2x	~1.3x-1.5x	↓	Lower lender comfort
Project IRR	~18-20%	~14-16%	↓	Reduced but acceptable returns
Payback Period	~7 years	~8-9 years	↑	Delayed recovery

Strategic interpretation:

- Most sensitive variable impacting overall financial performance
- Direct reduction in revenue leads to disproportionate decline in EBITDA (~50%)
- Highlights high dependency on maintaining price spread (Price – RM)
- Significant increase in break-even utilization reduces operational cushion
- DSCR weakens, indicating reduced lender comfort in downside scenario
- Emphasizes need for:
    - Long-term contracts
    - Product differentiation (printed / laminated bags)
- Reinforces importance of pricing power and market positioning

Scenario 2: 5% Increase in Raw Material Cost

A 5% increase in raw material cost impacts contribution margins but less severely than a selling price decline of equivalent magnitude. EBITDA reduces moderately, and break-even utilization increases. However, steady-state operations remain above break-even, and IRR remains within acceptable industrial benchmarks. The project retains financial viability under this moderate cost pressure scenario.

(AED,Million)

Particulars	Base Case	+5% RM Cost	Change	Strategic Comments
Sales Revenue	62.88	62.88	-	Revenue unaffected; cost-side pressure only
Raw Material Cost	(45.11)	(47.37)	(2.26)	Reflects polymer price volatility
Direct Expenses	(6.94)	(6.94)	-	No change
Contribution	10.83	8.57	(2.26)	Moderate decline
Indirect Expenses	(5.05)	(5.05)	-	Fixed overhead burden persists
EBITDA	5.78	3.52	(2.26)	Moderate impact (~39%)
EBITDA Margin (%)	9.2%	5.6%	↓	Manageable compression
Contribution / Ton (AED)	531	420	(111)	Unit economics remain viable
Break-Even Utilization (%)	57%	~65-68%	↑	Still within achievable range
DSCR (Steady-State)	~1.8x-2.2x	~1.4x-1.6x	↓	Adequate coverage maintained
Project IRR	~18-20%	~15-17%	↓	Still attractive
Payback Period	~7 years	~8 years	↑	Slight delay

Strategic Interpretation

- Reflects input cost volatility risk in polymer and scrap markets
- Moderate impact on profitability; less severe than price sensitivity

- Business remains operationally viable with positive margins
- Break-even increases but remains within manageable range
- DSCR remains above acceptable levels, indicating continued debt serviceability
- Key mitigation strategies:
  - Vendor diversification
  - Bulk procurement
  - Backward integration via recycling operations
- Demonstrates partial natural hedge due to integrated business model

Scenario 3: 1-Year Delay in Ramp-Up

A one-year delay in achieving projected ramp-up shifts revenue and EBITDA growth forward by one year, thereby postponing positive cash flow stabilization. While early-year liquidity stress increases and payback period extends, long-term project economics remain intact. The IRR declines moderately but stays within acceptable limits. This scenario primarily reflects execution risk rather than structural financial weakness.

Base Case vs Delayed Case (2026-2029 Comparison)

(AED,Million)

Particulars	Base Case	Delay Ramp-Up	Change	Strategic Comments
Sales Revenue	62.88	Timing shift	-	Revenue delayed, not reduced
Raw Material Cost	(45.11)	(45.11)	-	No structural change
Direct Expenses	(6.94)	(6.94)	-	Lower absorption initially
Contribution	10.83	Lower initially	↓	Temporary impact due to underutilization
Indirect Expenses	(5.05)	(5.05)	-	Fixed cost pressure early on
EBITDA	5.78	Lower initially	↓	Early-year suppression only
EBITDA Margin (%)	9.2%	Lower initially	↓	Temporary margin compression
Contribution / Ton (AED)	531	531	-	Unit economics unchanged
Break-Even Utilization (%)	57%	57%	-	No impact
DSCR (Steady-State)	~1.8x-2.2x	~1.8x-2.2x	-	Long-term stable
Project IRR	~18-20%	~16-17%	↓	Time value impact
Payback Period	~7 years	~8 years	↑	Delayed recovery

Strategic Interpretation

- Represents execution and implementation risk, not structural weakness
- Impacts timing of revenue and cash flow, not long-term profitability
- Early-stage DSCR and liquidity position may be temporarily stressed
- No change in:
  - Unit economics
  - Contribution margins
  - Break-even levels
- IRR and payback affected due to time value of money

- Key mitigation:
  - Strong project management
  - Vendor execution capability (Lohia)
- Pre-secured customer orders
- Long-term viability remains fully intact

Scenario 4: Combined Stress Case

The combined stress scenario represents a severe downside case where both selling price compression and raw material inflation occur simultaneously. This results in substantial margin erosion, significantly reduced EBITDA, and break-even levels approaching installed capacity limits. While the project may survive under short-term stress, prolonged combined adverse conditions would materially weaken financial stability unless corrective pricing or cost management actions are implemented.

(AED,Million)

Particulars	Base Case	Combined Stress	Change	Strategic Comments
Sales Revenue	62.88	59.73	(3.15)	Pricing pressure.
Raw Material Cost	(45.11)	(47.37)	(2.26)	Cost inflation.
Direct Expenses	(6.94)	(6.94)	-	Fixed cost burden.
Contribution	10.83	5.42	(5.41)	~50% drop.
Indirect Expenses	(5.05)	(5.05)	-	Fixed overhead stress.
EBITDA	5.78	0.37	(5.41)	Near breakeven.
EBITDA Margin (%)	9.2%	~0.6%	↓	Severe compression.
Contribution / Ton (AED)	531	~265	(266)	Weak economics.
Break-Even Utilization (%)	57%	~85-90%	↑	High risk zone.
DSCR (Steady-State)	~1.8x-2.2x	~1.0x-1.2x	↓	Tight servicing.
Project IRR	~18-20%	~12-14%	↓	Lower returns.
Payback Period	~7 years	>9 years	↑	Significant delay.

Strategic Interpretation

- Represents worst-case realistic scenario combining market and cost risks
- Severe compression in contribution and EBITDA (~near breakeven levels)
- Break-even utilization approaches 85–90%, significantly reducing cushion
- DSCR falls close to threshold, indicating tight debt servicing capacity
- Highlights critical dependency on:
  - Price-cost spread

- Operational efficiency
- Business remains marginally viable but financially stressed
- Requires strong management action:
    - Cost optimization
    - Price pass-through mechanisms
    - Focus on higher-margin product segments
- Reinforces importance of risk management and market diversification

Comparative Summary of Sensitivity Analysis

(AED,Million)

Particulars	Base	-5% Price	+5% RM	Delay	Combined
Revenue	62.88	59.73	62.88	Shifted	59.73
EBITDA	5.78	2.64	3.52	Lower early	0.37
Margin	9.2%	4.4%	5.6%	Lower early	~0.6%
BE Utilization	57%	~75%	~67%	57%	~90%
DSCR	~2.0x	~1.4x	~1.5x	~2.0x	~1.1x
IRR	~19%	~15%	~16%	~17%	~13%
Payback	~7 yrs	~8-9 yrs	~8 yrs	~8 yrs	>9 yrs

6.4 Risk Assessment and Mitigation

The proposed capacity expansion involves operational scale-up, market penetration, financial leverage, and regulatory compliance considerations. While the project demonstrates financial and technical viability, certain inherent risks must be evaluated and mitigated through structured management controls. The following section outlines key risk categories and corresponding mitigation measures.

Operational Risks

Operational risks primarily relate to commissioning delays, machinery performance, production ramp-up challenges, manpower capability, and maintenance management. The transition from a 6,000 MT facility to a 24,000 MT integrated recycling and woven packaging plant significantly increases operational complexity. Early-stage inefficiencies or technical underperformance could impact output, margins, and cash flow projections.

Risk Category	Risk Description	Impact	Mitigation Measures
Commissioning Delay	Delay in installation or commissioning of new woven line	Revenue delay, cash flow stress	Proven vendor selection (Lohia), installation supervision, phased ramp-up planning
Ramp-Up Risk	Slower-than-projected capacity utilization	Lower early EBITDA, DSCR pressure	Conservative ramp-up assumptions, active market linkage, gradual scaling
Machinery Breakdown	Unexpected equipment downtime	Production loss, repair cost	Preventive maintenance contracts, spare parts planning, local service support

Skilled Manpower	Insufficient technical expertise	Quality and efficiency issues	Operator training during commissioning, hiring experienced supervisors
Quality Control	Product rejection or inconsistent output	Margin erosion, customer dissatisfaction	In-house QC checks, process standardization, batch testing

Market & Financial Risks

Market and financial risks arise from pricing volatility, raw material cost fluctuations, demand variability, working capital intensity, and leverage exposure. The business operates in a commodity-influenced segment where margin sensitivity to selling price and polymer costs is high. Additionally, the project involves external debt of AED 14.4 million, creating initial years of debt servicing pressure.

Risk Category	Risk Description	Impact	Mitigation Measures
Selling Price Volatility	Competitive pricing pressure	Margin compression	Diversified product mix, contract-based customers, export expansion
Raw Material Price Fluctuation	Wastage sourcing cost	Contribution margin erosion	Multi-supplier sourcing, inventory planning, price pass-through strategy
Demand Slowdown	Industrial packaging demand decline	Lower utilization	Diversified customer base, GCC export focus
Debt Servicing Risk	Early-year DSCR pressure	Liquidity stress	Promoter funding support, structured repayment schedule
Working Capital Risk	Higher receivables/inventory	Cash flow strain	Credit control policies, inventory monitoring

Regulatory & Environmental Risks

The plastic recycling and packaging sector is subject to evolving environmental regulations, waste management frameworks, and sustainability compliance standards. While regulatory tightening may increase compliance requirements, it also creates structural demand for recycling and circular economy businesses. Non-compliance, however, could result in penalties or operational disruptions.

Risk Category	Risk Description	Impact	Mitigation Measures
Environmental Compliance	Stricter waste and emissions norms	Penalties, operational disruption	Compliance with UAE environmental standards, documented waste management
Regulatory Changes	New plastic regulations	Operational adjustments	Recycling-focused model aligned with circular economy policy
ESG Requirements	Large buyers requiring sustainability standards	Market access risk	Sustainable production positioning, recycled product portfolio
Licensing	Industrial approvals and renewals	Operational delays	Established presence in industrial zone, compliance documentation

Supply Chain Risks

Supply chain risks primarily relate to availability, quality, and pricing of plastic scrap feedstock and polymer inputs. Given the company's business model of waste procurement → pelletizing → woven production, uninterrupted feedstock supply is critical. Disruptions could impact production continuity and cost stability.

Risk Category	Risk Description	Impact	Mitigation Measures
Feedstock Availability	Shortage of plastic scrap	Production disruption	Diversified scrap sourcing (industrial + post-consumer)
Polymer Price Volatility	HDPE/LDPE price swings	Margin pressure	Strategic procurement planning, forward purchasing
Supplier Concentration	Dependence on limited suppliers	Supply instability	Multiple sourcing arrangements
Logistics Disruptions	Transport delays or cost increase	Delivery delays, cost escalation	Location advantage near highways and ports
Import Dependency	Reliance on imported raw materials	FX and supply risk	Regional sourcing strategy, inventory buffer

Overall Risk assessment

The project entails moderate operational and financial risk during the initial ramp-up phase, primarily due to capacity absorption timelines, debt servicing commitments, and execution dependencies. However, the long-term structural outlook remains strong, supported by scalable capacity, improving operating leverage, and steady-state cash flow generation. The key risks identified are measurable and can be effectively managed through structured operational planning, selection of reliable machinery vendors, diversified customer outreach, and disciplined financial oversight. In particular, robust vendor management for raw material procurement-leveraging the company's existing trading experience and supplier network-will play a critical role in mitigating margin volatility and financial exposure in an evolving polymer market environment. This integrated sourcing capability strengthens resilience against price fluctuations and enhances overall project stability.

Strategic Recommendations and Evaluation of Proposed Capacity Expansion

The proposed expansion from 6,000 MT to 24,000 MT represents a transformational shift in the company's scale, operational capability, and market positioning. The current facility operates near optimal utilization levels, with limited scope for incremental revenue growth under the existing infrastructure. The expansion is therefore strategically aligned not only with growth ambitions but also with structural necessity to unlock operating leverage and improve financial sustainability.

The integration of tape extrusion, weaving, coating, printing, and converting lines positions the company as a vertically integrated packaging manufacturer rather than a standalone recycler or trader. This shift enhances value addition at product level as well as develop capability for unserved customer groups for growth perspectives.

Strategically, the expansion enables:

- Transition from small-scale recycling to integrated woven packaging manufacturing
- Better fixed cost absorption and improved EBITDA margins
- Diversification of revenue streams (pellets + woven sacks + films)
- Entry into industrial packaging segments such as cement, fertilizer, and polymer bags
- Strengthened competitiveness in the UAE and GCC markets

Technical Evaluation

From a technical perspective, the proposed machinery suite (LOHIA-based integrated woven line) is appropriate for the intended production scale and product mix. The selected equipment offers:

- Proven industrial performance
- Energy efficiency
- Strong service support within the UAE
- Scalability for future capacity optimization

The plant layout design supports linear production flow, minimizing material handling inefficiencies. The integration of upstream recycling with downstream woven conversion enhances supply control and reduces dependency on external granule suppliers.

The technical configuration is therefore considered robust and compatible with the project's long-term industrial objectives.

Financial Evaluation

The financial assessment indicates that the expansion is economically feasible under base-case assumptions.

Key Observations:

- Project IRR: ~19-21%
- Equity IRR: >30%
- Payback Period: 6-7 years
- Break-even Utilization: ~58-60%
- Steady-State DSCR: >1.5x

While the initial 2-3 years reflect ramp-up and debt servicing pressure, steady-state operations demonstrate strong cash generation and margin stabilization. Sensitivity analysis confirms that the project remains viable under moderate stress scenarios, with risk increasing only under combined adverse pricing and cost conditions.

The capital structure, including AED 14.4 million external debt and AED 4.0 million promoter contribution, appears balanced, with adequate long-term repayment capacity post-stabilization.

Conclusion

Based on technical, financial, operational, and market evaluation, the proposed capacity expansion is strategically potential and financially viable. The project demonstrates:

- Strong long-term return potential
- Acceptable leverage structure
- Adequate break-even cushion
- Structural alignment with regional demand trends
- Manageable and identifiable risk profile

Although moderate operational and financial risk exists during the ramp-up phase, the long-term economic fundamentals remain robust. With disciplined execution and prudent financial management, the expansion is expected to significantly enhance the company’s scale, profitability, and market position.

Disclaimer: The financial projections have been prepared based on the unaudited FY2025 financial statements as shared, which have been treated as actual results for the purpose of this analysis. Several supporting documents were not made available for review; accordingly, where sufficient verification was not possible, assumptions have been applied based on secondary research, industry benchmarks, management inputs, historical performance trends, reasonable market expectations and professional judgment consistent with consulting engineering practices. As such, these projections are indicative in nature and remain subject to revision upon receipt of audited financial statements or additional substantiating information.

List of Annexures

1. Excel back-up of calculations, assumptions and financial summary

List of images